

ZFT is the major iron and zinc transporter in *Toxoplasma gondii*

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This **important** study identifies a putative iron and zinc transporter in the plasma membrane of the obligate intracellular pathogen, *Toxoplasma gondii*. Using an array of different approaches, the authors **convincingly** demonstrate that this transporter regulates diverse cellular processes, including parasite metabolism and differentiation. This work will be of broad interest to cell biologists and biochemists studying metal ion transport mechanisms.

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Abstract

Transition metals, such as iron and zinc, are indispensable trace elements for eukaryotic life, acting as co-factors in essential processes ranging from respiration and metabolism to DNA replication. These metals can be transported into cells by an evolutionary-conserved family of metal transporters; however, how the ubiquitous mammalian parasite *Toxoplasma gondii* acquires essential metals has been unknown. Here, we have identified and characterised the first iron and zinc importer in *T. gondii*. This transporter, named ZFT, localised to the parasite plasma membrane and is essential for the parasite's life cycle. We find ZFT is regulated by iron availability and overexpression sensitises cells to excess iron and zinc. Using a conditional knockdown system, we find that knockdown of ZFT leads to reduction in mitochondrial respiration and a switch to a more quiescent lifecycle stage. To confirm transport activity, we find that knockdown of ZFT leads to a reduction in parasite-associated zinc and iron, and ZFT expression complements loss of zinc transporter activity in a yeast model. Overall, we have identified the first metal uptake transporter in *T. gondii* and demonstrated the importance of iron and zinc to the parasite. This finding advances our understanding how this obligate intracellular parasite acquires nutrients from its host.

Introduction

Iron, and other transition metals such as zinc, manganese and copper, are essential nutrients for almost all life, playing vital roles in biological processes including DNA replication, translation, cellular respiration and metabolism (Teh et al., 2024 [DOI](#)). As metals cannot diffuse across membranes, organisms have developed a range of strategies to take up metals, e.g. through endocytosis of iron-bound transferrin by mammalian cells, or siderophore uptake systems in bacteria. However, iron and other transition metals can catalyse reactive oxygen species, and so intracellular levels are tightly regulated (Galaris et al., 2019 [DOI](#); Imlay et al., 1988 [DOI](#)).

Toxoplasma gondii is an obligate intracellular apicomplexan parasite which chronically infects around 30% of the world's population (Bisetegn et al., 2023 [DOI](#)). Although typically asymptomatic, Toxoplasmosis can be dangerous in immunocompromised individuals and to the developing foetus, resulting in severe congenital defects (Desmonts and Couvreur, 1974 [DOI](#); Wong and Remington, 1994 [DOI](#)). Unusually among the Apicomplexa, *T. gondii* is promiscuous, capable of infecting all nucleated cells (Tenter et al., 2000 [DOI](#)) and all warm-blooded species, causing significant losses in sheep and goats (Stelzer et al., 2019 [DOI](#)). Iron is essential for the parasite where it is used in essential cellular processes, including haem biosynthesis (Bergmann et al., 2020 [DOI](#); Kloehe et al., 2021 [DOI](#)) and iron-sulphur (Fe-S) cluster biogenesis pathways (Aw et al., 2021 [DOI](#)) and depletion of iron using chelators restricts parasite replication *in vitro* and *in vivo* (Aghabi et al., 2023 [DOI](#); Ferrer et al., 2012 [DOI](#); Mahmoud, 1999 [DOI](#); Raventos-Suarez et al., 1982 [DOI](#); Sloan et al., 2025 [DOI](#)). Beyond iron, the role of other transition metals in *Toxoplasma* biology is less well understood. It is predicted at least 800 *Toxoplasma* proteins require zinc, several of which have already been shown to be essential (Gissot et al., 2017 [DOI](#); Semenovskaya et al., 2020 [DOI](#)) and loss of zinc storage leads to mild parasite defects (Chasen et al., 2019 [DOI](#)). However, the transport pathways, use of zinc and effects of zinc depletion have not been investigated. Further, although both iron and zinc are essential for *Toxoplasma*, the mechanisms of their uptake in *Toxoplasma gondii* have not yet been established.

The ZIP (Zrt-/Irt-like protein)-domain containing proteins represent a large, highly conserved family of metal transporters (Guerinot, 2000 [DOI](#)). ZIP-domain proteins mediate both iron and zinc uptake and intracellular trafficking in mammalian, plant and fungal cells. ZIP-domain containing proteins have also been utilised in eukaryotic parasites; a *Leishmania* ZIP-domain containing protein was identified as the major iron importer and was essential for intracellular replication (Huynh et al., 2006 [DOI](#); Jacques et al., 2010 [DOI](#)). In *Plasmodium*, the plasma membrane-localised ZIP-domain containing protein ZIPCO was shown to be essential only in the liver stage of the parasite and was identified as a potential iron and zinc transporter based on partial rescue of a growth phenotype with iron and zinc (Sahu et al., 2014 [DOI](#)). In the blood stages, iron appears to be transported from the *Plasmodium* food vacuole to the cytosol by DMT1, where it could be utilised by the apicoplast and mitochondrion (Loveridge and Sigala, 2024 [DOI](#); Zhong and Zhou, 2023 [DOI](#)).

Toxoplasma gondii encodes four predicted ZIP-domain containing proteins: TGME49_261720, TGME49_254080, TGME49_500163 and TGME49_225530, none of which have been previously characterised. Our previous work identified that TGME49_261720 mRNA was stabilised under low iron by the presence of a stem-loop structure in the 3' UTR (Sloan et al., 2025 [DOI](#)). Loss of this regulation inhibited parasite survival under low iron, however the function of the protein was not investigated. Here, we characterise TGME49_261720, named Zn and Fe Transporter (ZFT), in detail. We find ZFT localisation and abundance is dynamic and dependent on parasite replication, vacuolar stage and iron availability. ZFT is essential for the parasite's lytic lifecycle and knockdown leads to inhibition of mitochondrial respiration, changes to apicoplast biogenesis and initiation of stage conversion. Unlike in *Plasmodium*, ZFT knockdown cannot be rescued by addition of exogenous metals. By quantifying parasite-associated metals, we show ZFT knockdown

parasites have significantly less iron and zinc, and expression of ZFT rescues zinc uptake in mutant yeast. These data strongly support the role of ZFT as the major iron and zinc transporter in *Toxoplasma gondii*.

Results

Iron transport in *Toxoplasma gondii*

We examined the genome of *Toxoplasma* and identified four genes encoding predicted ZIP-domains (Amos et al., 2022 [\[1\]](#)). Of these, both TGME49_254080 and TGME49_500163 were predicted to contain insufficient transmembrane domains (6 and 3 respectively, rather than 8 or 9) and the ZIP domain of TGME49_254080 was relatively poorly conserved and were not considered further. Both TGME49_261720 and TGME49_225530 encoded ZIP-domains with high predicted confidence (E value < 7×10^{-30}), are predicted to have 8 transmembrane domains and share limited sequence homology with PfZIPCO (21% and 26% identity respectively). We examined the sequence conservation at the binuclear metal centres (BNC) M1 and M2 in transmembrane domains $\alpha 4$ and $\alpha 5$, which have been shown to be required for metal binding in a bacterial ZIP homolog (BpZIP) (Wu et al., 2022 [\[2\]](#)). At M1, *Toxoplasma* TGME49_261720 contains an unusual HGxxEG motif, which we found only in coccidian ZIPs and in some marine organisms (e.g. *Branchiostoma floridae*), the impact of which is not clear (Figure 1a [\[3\]](#)). In contrast, TGME49_225530 contains a HAxEG motif, also found in prokaryotic ZIP9-like proteins (Hu, 2021 [\[4\]](#)). Interestingly, all parasite ZIP-domain proteins examined have a HK motif at the M2 metal binding site (Figure 1a [\[3\]](#)), characteristic of the ZIP family of ZIP-domain proteins, shared in HsZIP1 and HsZIP2 (Dufner-Beattie et al., 2003 [\[5\]](#)). This lysine likely prevents metal binding at this site, however has been shown to be required for transport activity in HsZIP2, possibly with a role in pH-responsivity of the transporter (Gyimesi et al., 2019 [\[6\]](#)). Based on the unusual conserved motif, the predicted plasma membrane localisation, essentiality in tachyzoites and our previous results demonstrating iron-mediated regulation (Sloan et al., 2025 [\[7\]](#)), we decided to functionally characterise the role of TGME49_261720. We constructed a structural prediction of TGME49_261720 as a dimer (in common with other reported ZIP-domain containing proteins). TGME49_261720 had 8 transmembrane domains with a large, unstructured loop between helices 3 and 4, predicted to be on the outer lumen, the role of which is unclear. The highly conserved metal binding residues H314 and E318 are indicated within the core of the predicted sequence (Figure 1b [\[3\]](#)).

ZFT localisation is dynamic and vacuolar stage dependent

We have previously shown that the 3' UTR of ZFT plays an important role in iron-mediated regulation (Sloan et al., 2025 [\[7\]](#)) and so we tagged ZFT with a 3xHA epitope tag with the endogenous 3' UTR (ZFT-3HA_{ZFT}) as previously described (Sloan et al., 2025 [\[7\]](#)) (Figure 1c [\[3\]](#)). Western blot demonstrated the protein ran at the expected size of around 50 kDa, although with multiple, non-specific lower bands (asterisks) (Figure 1d [\[3\]](#)). Using immunofluorescence, we found that ZFT localised to the periphery of the parasite with a second foci at the basal pole of the cell (Figure 1e [\[3\]](#)). To determine if changing the UTR affected ZFT localisation, we also tagged it with a 3xHA epitope tag followed by the *sag1* 3' UTR (ZFT-3HA_{SAG1}) as previously described (Sloan et al., 2025 [\[7\]](#)) (Figure S1a [\[3\]](#)) and saw a similar localisation (Figure S1b [\[3\]](#)). Interestingly, localisation was highly dependent on the number of parasites per vacuole. In early stages (2-4 parasites/vacuole) ZFT localised peripherally, however this shifted to basal localisation in larger vacuoles (>8 parasites) (Figures 1e, f [\[3\]](#)). Accompanying this change in localisation, we quantified the amount of ZFT-3HA_{ZFT} and found that the mean fluorescence intensity (MFI) of ZFT/parasite was significantly ($p < 0.001$, one-way ANOVA, Tukey corrected for multiple comparisons) higher in vacuoles with fewer parasites, compared to larger vacuoles with 8 parasites (Figure 1g [\[3\]](#)). This suggests that ZFT is degraded in a vacuolar stage-dependent manner. Previously, we showed that ZFT levels are regulated by low iron (Sloan et al., 2025 [\[7\]](#)). To determine if the localisation of ZFT

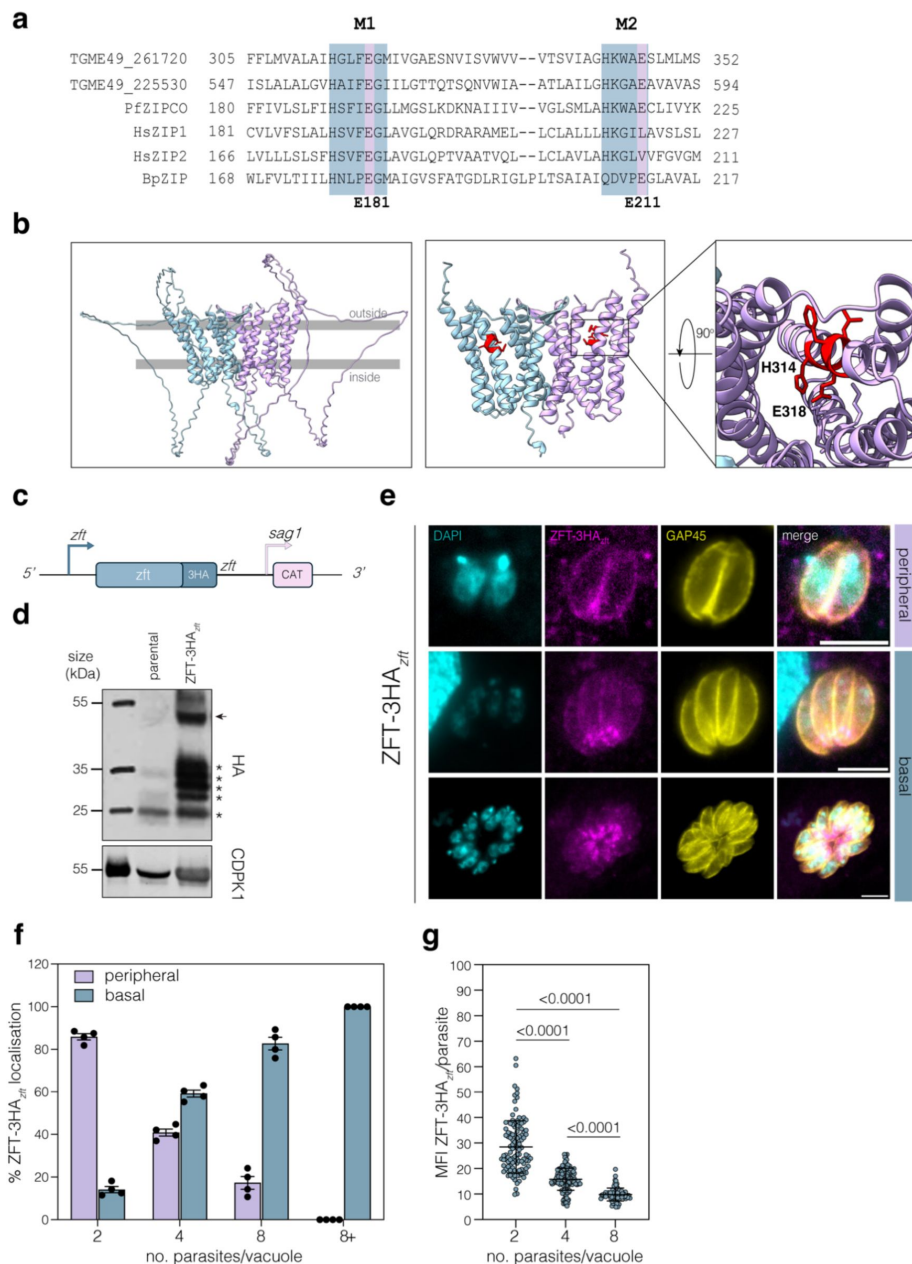


Figure 1.

ZFT is a ZIP-domain containing protein with dynamic localisation.

a. Alignment of ZIP proteins from *T. gondii* (TGME49_261720), *T. gondii* (TGME49_266800), *P. falciparum* (PF3D7_1022300), HsZIP1 and HsZIP2. Key conserved residues in the binuclear metal centres (BNC) M1 and M2 in transmembrane domains a4 and a5, which have been shown to be required for metal binding are highlighted. Conserved glutamate residues highlighted in purple. **b.** AlphaFold model of TGME49_261720 dimer with key residues highlighted. **c.** Schematic of F3ΔHX ZFT-3HA_{zft} strain construction to endogenously tag ZFT with 3xHA epitope tags at the C-terminal under the endogenous promoter with the endogenous 3' UTR. **d.** Western blot analysis of ZFT-3HA_{zft} confirming the expected band size (50 kDa). Non-specific bands indicated with asterisks. CDPK1 (TGME49_301440) used as a loading control. **e.** Immunofluorescence of ZFT-3HA_{zft} demonstrating dynamic peripheral or basal localisation. Scale bars 5 μm. **f.** Percentage of ZFT-3HA_{zft} localisation (peripheral or basal) in respect to number of parasites per vacuole. 100 parasites were counted for each condition. Bars at mean of four independent experiments, ± SD. **g.** Mean fluorescence intensity (MFI) of ZFT-3HA_{zft} per parasite in relation to the number of parasites per vacuoles. Each point represents individual MFI/parasite values from three independent experiments. Bars at mean of the experiments ± SD. *p* values from ordinary one-way ANOVA, Tukey corrected for multiple comparisons.

was affected under iron starvation, we treated the parasites with the iron chelator deferoxamine (DFO) for 24 h. Under low iron conditions, ZFT-3HA_{zft} localisation was maintained at the parasite periphery, although some intracellular vesicles could be observed (Figure S1c [↗](#)).

These data show ZFT expression and localisation is dynamic depending on vacuolar stage.

ZFT expression is regulated by excess iron

The localisation and expression of transporters is often regulated by substrate availability (Conte and Walker, 2011 [↗](#); Gao and Dubos, 2021 [↗](#); Lee et al., 2020 [↗](#)), and we have previously shown the iron transporter VIT is regulated by iron in *Toxoplasma* (Aghabi et al., 2023 [↗](#)). We have previously demonstrated the *zft* 3' UTR is sufficient to stabilise the mRNA under low iron (Sloan et al., 2025 [↗](#)), however expression under excess iron was not examined. To determine if ZFT-3HA_{zft} is regulated by excess iron, we treated parasites with exogenous ferric ammonium chloride (FAC). At 18 h post treatment, ZFT-3HA_{zft} levels were slightly (to around 87% of the untreated) but significantly ($p = 0.028$, two-tailed paired t test) reduced (Figures 2a, b [↗](#)). However, by 24 h post treatment we found that ZFT-3HA_{zft} expression was reduced to around 40% of the untreated ($p = 0.0041$, two-tailed paired t test) (Figures 2c, d [↗](#)). We confirmed loss of ZFT-3HA_{zft} expression under excess iron by immunofluorescence (Figures 2e, f [↗](#)), demonstrating ZFT expression is significantly reduced under excess iron.

Overexpression of ZFT leads to a change in iron sensitivity

Initial attempts to overexpress ZFT were unsuccessful, suggesting that overexpression was toxic. However, we were able to stably overexpress ZFT by coupling a second copy of ZFT-Ty expressed from a strong promotor, pTUB8 (Fr  nal et al., 2010 [↗](#)) with DHFR using a P2A-slip peptide (Markus et al., 2019 [↗](#)) and maintaining the line in pyrimethamine (Figure 2g [↗](#)). We validated the presence of second copy by PCR (Figure S2a [↗](#)), and the protein by western blotting (Figure 2h [↗](#)). To determine the effect of overexpression of ZFT on parasite fitness, we performed a plaque assay of both parental and ZFT-Ty parasites (Figure 2i [↗](#)). Parasites overexpressing ZFT form significantly ($p = 0.0027$, unpaired two-tailed t test) fewer plaques than parental parasites (Figure 2j [↗](#)), with significantly ($p < 0.0001$, two-tailed unpaired t test with Welch's correction) smaller area (Figure 2k [↗](#)), confirming that overexpression of ZFT inhibits parasite replication. Overexpression of metal transporters can alter the sensitivity to environmental metal conditions (Connolly et al., 2002 [↗](#); Li and Kaplan, 1998 [↗](#)). To test this, we quantified the effects of ZFT overexpression on iron sensitivity. Our data show parasites overexpressing ZFT-Ty are significantly ($p = 0.0027$, two-tailed unpaired t test) more sensitive ($EC_{50} = 0.386$ mM, 95% C.I. 0.315 to 0.479 mM) to excess iron, compared to the parental line ($EC_{50} = 1.29$ mM, 95% C.I. 0.813 to 2.01 mM) (Figures 2l [↗](#), S2b). Excess iron leads to parasite death largely through oxidative stress (Aghabi et al., 2023 [↗](#)). To confirm that parasites were not more sensitive based on non-specific increased sensitivity to oxidative stress, we treated parasites with sodium arsenite (Ars), which has previously been used to induce oxidative stress in *T. gondii* (Aghabi et al., 2023 [↗](#); Augusto et al., 2021 [↗](#)). However, overexpression of ZFT does not lead to any significant change in oxidative stress sensitivity (Figure S2c, d [↗](#)), suggesting the phenotype is specific to iron. We then tested how ZFT-Ty overexpression affected sensitivity to iron starvation using DFO. We found parasites overexpressing ZFT are significantly ($p = 0.0246$, two-tailed unpaired t test) more resistant ($EC_{50} = 22.81$ μ M, 95% C.I. 16.8 to 30.6 μ M) to iron chelation, compared to the parental line ($EC_{50} = 8.18$ μ M, 95% C.I. 6.94 to 9.71 μ M) (Figures 2m [↗](#), S2e). These data suggest that overexpression of ZFT leads to alterations in parasite iron sensitivity.

ZFT is essential for parasite survival

To determine the importance of ZFT in parasite replication, an inducible knockdown line was generated by replacing the promotor of ZFT with the T7S4 promotor which is inhibited in the presence of ATc (Sheiner et al., 2011 [↗](#)) (Figure 3a [↗](#)), which was validated by PCR (Figure S3a [↗](#)).

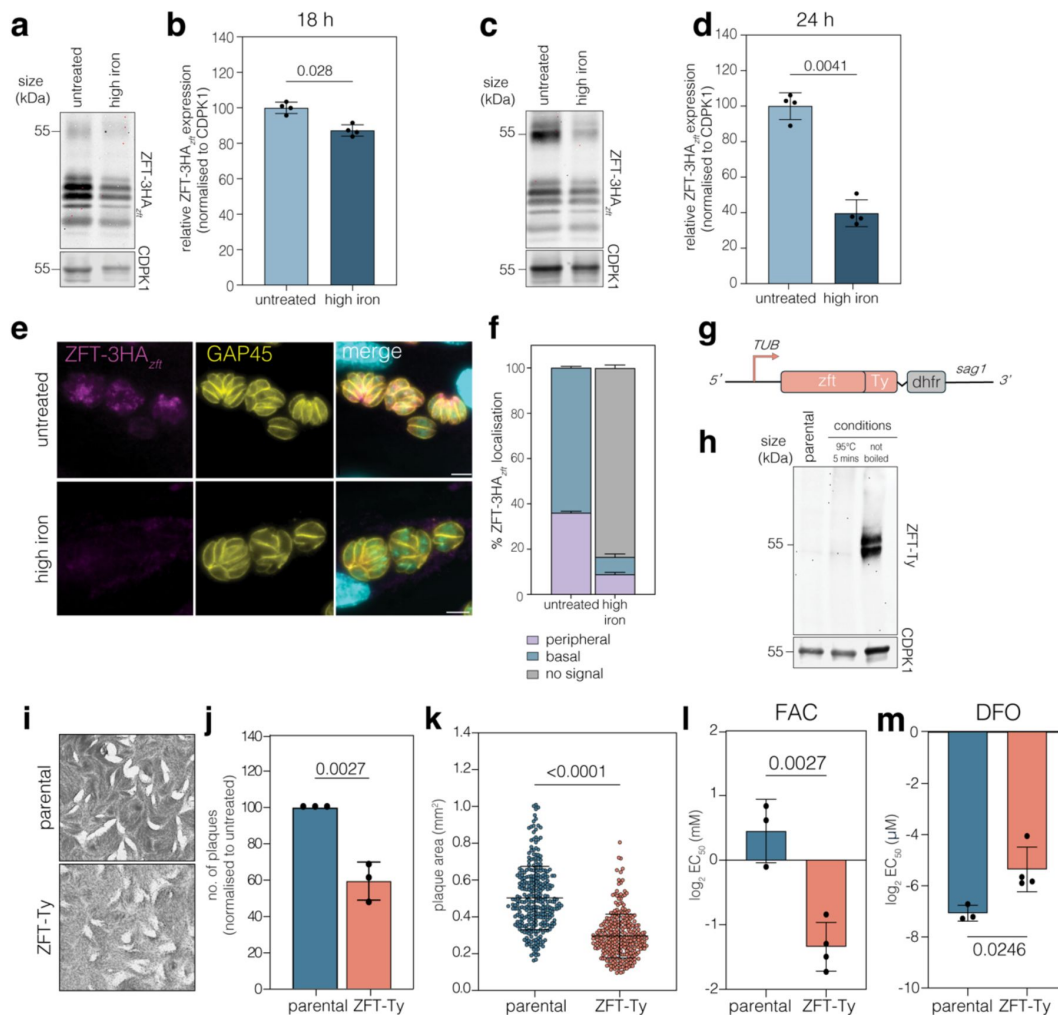


Figure 2.

ZFT-3HA_{zft} is regulated by iron, and overexpressing ZFT leads to a change in iron sensitivity.

a. Western blot analysis showing ZFT-3HA_{zft} expression levels 18 h post infection in untreated and high iron (500 μ M FAC) conditions. CDPK1 was used as a loading control. **b.** Quantification of ZFT-3HA_{zft} from three independent experiments. Bars at mean $\pm$ SD. p values from two-tailed paired t test. **c.** As above, at 24 h post infection in untreated and high iron (500 μ M FAC) conditions. **d.** Quantification of ZFT-3HA_{zft} from three independent experiments. Bars at mean $\pm$ SD. p values from two-tailed paired t test. **e.** Immunofluorescence assay of ZFT-3HA_{zft} at 24 h post infection in untreated and high iron (500 μ M FAC) conditions. Scale bars 5 μ m. **f.** Quantification of ZFT-3HA_{zft} localisation (peripheral/basal or no signal) in untreated and high iron conditions. Bars at mean of three independent experiments, $\pm$ SD. **g.** Schematic of ZFT-Ty strain construction to overexpress ZFT from the TUB8 promoter in RHtdTomato parental line. **h.** Western blot of ZFT-Ty showing expected size. **i.** Plaque assay of parental and ZFT-Ty parasites in untreated conditions. Quantification of plaque number (**j**) and area (**k**) for parental and ZFT-Ty parasites, p values are unpaired two-tailed t test. Points represent individual plaque areas from three independent experiments, bars at mean $\pm$ SD. p values from two-tailed unpaired t test with Welch's correction. Graphs showing mean parasite EC₅₀ for FAC (**l**) and DFO (**m**) for RHtdTomato and ZFT-Ty parasites. Each point represents an independent experiment, bars at mean of $n=3$, $\pm$ SD. p values from two-tailed unpaired t test.

To verify conditional knockdown, we performed RT-qPCR as previously described (Sloan et al., 2025). By 24 h post induction with ATc, we found that *zft* mRNA was significantly ($p = 0.0032$, two-tailed unpaired *t* test with Welch's correction) reduced (Figure 3b). We epitope tagged ZFT in the T7S4-ZFT background (Figure 3c) and confirmed integration by PCR (Figure S3b). Western blotting demonstrated that swap of the promotor led to a significant (around 5 times) overexpression of ZFT in the T7S4-ZFT cell line, compared with the endogenous promotor (Figure S3c, d). We validated the ATc-induced inhibition of expression by western blot (Figure 3d) and show that there is a significant ($p < 0.0001$, two-tailed paired *t* test) drop in ZFT-3HA_{sag1} expression levels by 24 h post induction. At 48 h after adding ATc, only around 10% of the protein remained ($p = 0.0131$, two-tailed paired *t* test) (Figure 3e). Moreover, we found that ZFT-3HA_{sag1} was undetectable by immunofluorescence by 48 h post induction (Figure 3f). Previous studies on both bacterial and eukaryotic ZIPs suggest that these proteins function in either homodimers or heterodimers (Ahern et al., 2020; Bin et al., 2011; Lin et al., 2010; Taylor et al., 2016). To determine if ZFT-3HA_{sag1} forms a complex in *Toxoplasma*, we performed a Blue native PAGE (BN-PAGE) and found that ZFT-3HA_{sag1} forms a complex of ~100 kDa, which is lost upon knockdown, suggesting the formation of a dimer (Figure 3g).

To examine the role of ZFT, a plaque assay was performed (Figure 3h). Replacement of the ZFT promotor alone led to a significant ($p = 0.0052$, two-way ANOVA, Tukey corrected for multiple comparisons) defect in plaque number and area (Figures 3h, i, j). This, combined with our previous study (Sloan et al., 2025), demonstrates that correct expression of ZFT is important for parasite fitness. Addition of ATc renders T7S4-ZFT parasites unable to form plaques in the host cell monolayer (Figures 3h, i, j), confirming that expression of ZFT is essential for the parasite. In *Plasmodium*, the growth defect upon loss of DMT1 (Loveridge and Sigala, 2024) or ZIPCO (Sahu et al., 2014) was able to be rescued by addition of exogenous substrate. To determine if loss of ZFT could be rescued by addition of iron, we treated ZFT knockdown cells with 200 μ M FAC and quantified growth (Figure 3k). We found that replacing the ZFT promoter alone led to a significant reduction in plaque area ($p < 0.0001$, two-tailed unpaired *t* test with Welch's correction) in high iron conditions (200 μ M FAC) compared to untreated conditions (Figure 3l), potentially due to the overexpression of ZFT under the T7S4 promotor. However, exogenous iron was unable to rescue the growth defect, either by plaque number or area (Figures 3k, m, n). To determine if iron could restore growth when some ZFT remained, we also performed a plaque assay with half the standard concentration of ATc (0.25 μ M) (Figure S3e). We find that in these conditions, ZFT KD parasites are still unable to form any plaques, even in the presence of excess iron (200 μ M FAC) (Figures S3e, f, g). This suggests a strong, irreversible growth defect and the essentiality of ZFT as the major iron transporter.

ZFT knockdown alters organellar function

Iron is required for proteins essential in parasitic organelles such as the apicoplast and mitochondrion (Maclean et al., 2024; Pamukcu et al., 2021; Renaud et al., 2022) and disruption of organellar function can lead to morphological changes (Mallo et al., 2021; Wu et al., 2022). To determine if ZFT knockdown was associated with changes in the apicoplast, we initially imaged at 24 h and 48 h of ATc, however saw no clear morphological defects (Figure 4a). To determine if there is a defect in apicoplast protein import post-ZFT depletion, we examined CPN60 processing at 48 h treatment and saw no change (Figure S4a). We also examined protein lipoylation, as the apicoplast contains the essential Fe-S-protein LipA, which is required for lipoylation of the apicoplast-localised pyruvate dehydrogenase E2 (Pamukcu et al., 2021; Crawford et al., 2006). Using an anti-lipoic acid antibody, we were able to detect lipoylated PDH-E2 at 48 h post ATc treatment, suggesting that at this time point, apicoplast functions are maintained (Figure 4b). Given the previously established delayed-death phenotype of the apicoplast, we determined the effects of ZFT knockdown on apicoplast maintenance after a further round of mechanical release and invasion. Here, we observed a small but significant increase ($p = 0.04$, two way ANOVA, Tukey corrected for multiple comparisons) in elongated apicoplasts (Figure 4c, d). We also quantified parasite replication and found a

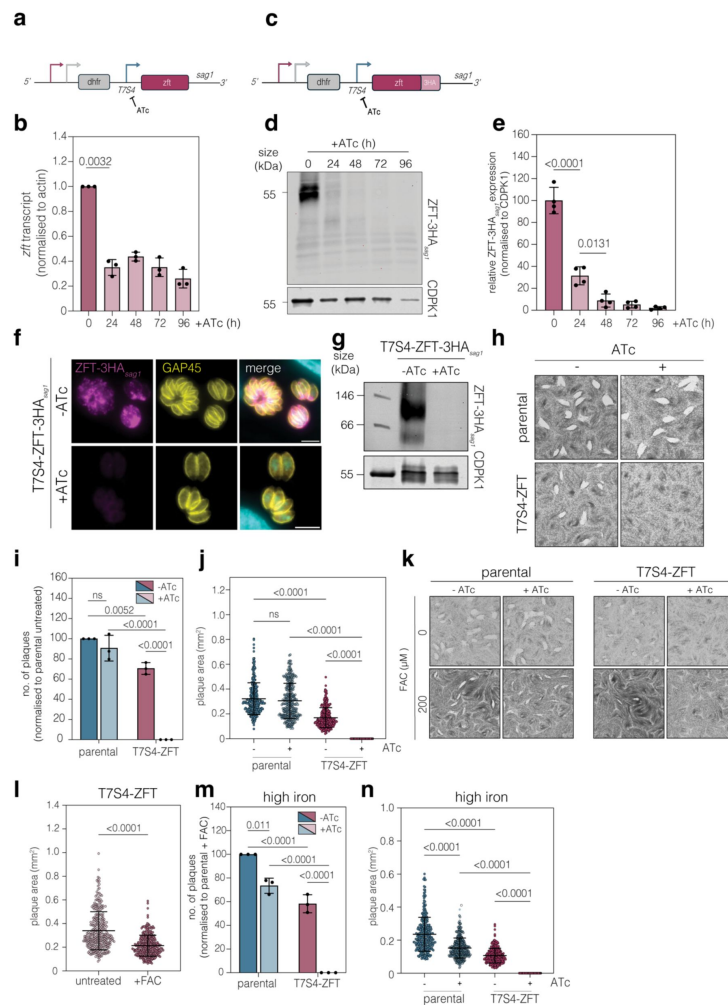


Figure 3.

ZFT knockdown blocks parasite replication.

a. Schematic of F3ΔHX T7S4-ZFT under the T7S4 inducible promoter. **b.** RT-qPCR showing ZFT mRNA abundance, relative to 0 h, normalised to actin. Bars at mean of three independent experiments, $\pm$ SD. p values from two-tailed unpaired t test. **c.** Schematic of F3ΔHX T7S4-ZFT-3HA_{sag1} under the T7S4 inducible promoter. **d.** Western blot demonstrating successful knockdown of ZFT at 24, 48, 72 and 96 h post ATc induction. CDPK1 used as a loading control. **e.** Quantification of (d), relative to 0 h, normalised to CDPK1. Bars at mean of four independent experiments, $\pm$ SD. p values from two-way ANOVA, Tukey corrected for multiple comparisons. **f.** Immunofluorescence assay showing T7S4-ZFT-3HA_{sag1} expression in untreated parasites and after induction with ATc for 48 h. Scale bar 5 μ m. **g.** Blue native PAGE of T7S4-ZFT-3HA_{sag1}, untreated and + ATc 48 h induction, showing a ZFT-HA complex at around 100 kDa. CDPK1 used as a loading control. **h.** Plaque assay of parental and T7S4-ZFT parasites untreated and ATc treated, 6 days, showing knockdown of ZFT renders the parasite unable to form plaques in the host cell monolayer. **i.** Quantification of (h) showing number of plaques, normalised to parental untreated. Bars at mean of three independent experiments, $\pm$ SD. p values from two-way ANOVA, Tukey corrected for multiple comparisons. **j.** Quantification of plaques area from (h). Points represent individual plaques from three independent experiments. Bars at mean $\pm$ SD. p values from ordinary one-way ANOVA, Tukey corrected for multiple comparisons. **k.** Plaque assays with addition of 200 μ M FAC does not rescue ZFT knockdown parasites. **l.** Quantification of plaque area generated by T7S4-ZFT parasites in untreated and high iron (200 μ M FAC) conditions. Points represent individual plaque areas from three independent experiments. Bars at mean $\pm$ SD. p values are from two-tailed unpaired t test with Welch's correction. **m.** Quantification of plaque number, normalised to parental untreated. Bars at mean of three independent experiments, $\pm$ SD. p values from two-way ANOVA, Tukey corrected for multiple comparisons. **n.** Quantification of plaque area generated by parental and T7S4-ZFT parasites with and without ATc, in the presence of excess iron (200 μ M FAC). Points represent individual plaque areas from three independent experiments. Bars at mean $\pm$ SD. p values from ordinary one-way ANOVA, Tukey corrected for multiple comparisons.

significant ($p < 0.0055$ at the 1, 2, 4, and 8 parasites/vacuole, two way ANOVA with Šidák correction) decrease in parasites/vacuole in the second round of invasion, compared to untreated (Figure 4e).

Iron is also required within the parasite mitochondrion (Maclean et al., 2024; Pamukcu et al., 2021). We first examined gross parasite mitochondrial morphology by light microscopy (Figure 4f) and saw no significant alterations. To assess if ZFT knockdown affects mitochondrial respiration, oxygen consumption rate was quantified using a Seahorse assay (Hayward et al., 2022) (Figure 4g). We found that 48 h of ZFT knockdown significantly reduced basal (Figure 4h) and maximal (Figure 4i) mitochondrial oxygen consumption rates (mOCR) ($p = 0.0233$, $p = 0.0054$ respectively, two-tailed unpaired t test with Welch's correction), compared to untreated parasites. Interestingly, we find ZFT knockdown does not significantly change the extracellular acidification rate (ECAR), a proxy for glycolysis (Hayward et al., 2022) (Figure S4b) and the metabolic map suggests that ZFT knockdown leads to a reduction in oxidative phosphorylation and a more quiescent phenotype (Figure 4j).

Within the mitochondrion, iron is incorporated in proteins of the electron transport chain (Maclean et al., 2021). A decrease in mitochondria respiration could be caused by a depletion of proteins in the electron transport chain or the activity of complexes within the electron transport chain. To determine how these are affected by depletion of ZFT, we firstly examined the expression of the key mitochondrial protein succinate dehydrogenase B (SDHB). SDHB has essential roles both in the TCA cycle and in complex III (Maclean et al., 2021; Silva et al., 2023) and contains three Fe-S clusters. We endogenously tagged SDHB (TGME49_215280) in the T7S4-ZFT background (Figure S4c), confirmed integration by PCR (Figure S4d), and quantified protein levels following 48 h of ZFT knockdown. Upon depletion of ZFT, SDHB-3HA expression levels were significantly ($p = 0.0205$, two-tailed paired t test) reduced to around 34%, compared to untreated cells (Figure 4k, l). As a control, we also tagged the essential cytosolic Fe-S protein ABCE1 (TGME49_216790) (Maclean et al., 2024) (Figure S4e) and confirmed integration by PCR (Figure S4f). However, ABCE1-3HA expression levels remained unchanged at 48 h upon ZFT knockdown ($p = 0.1758$, two-tailed paired t test) (Figure S4g, h), confirming the lack of a global effect on Fe-S proteins upon ZFT knockdown. We also examined the activity of complex IV, which is predicted to contain both Fe-S proteins and haemoproteins (Leonard et al., 2023), by in gel activity assay (Lacombe et al., 2019; Silva et al., 2023). We find that ZFT knockdown reduced complex IV activity at 48 h post ZFT knockdown (Figure 4m). Changes in mitochondrial respiration have previously been linked to changes in cristae morphology (Baker et al., 2019; Huet et al., 2018). Using transmission electron microscopy (TEM) at 72 h post ATc treatment, we saw no clear changes in cristae morphology, compared to untreated parasites (Figure S4i).

These data demonstrate ZFT knockdown inhibits apicoplast replication and reduces mitochondrial respiration, potentially due to loss of iron-containing proteins in the mitochondrial electron transport chain.

Knockdown of ZFT triggers partial parasite differentiation

To determine the effect of ZFT knockdown, we examined parasite replication. Interestingly, replication was not immediately inhibited, and at 48 h post ATc addition we observed large vacuoles containing many nuclei (Figure 5a), although these large vacuoles did not egress from the host. Around 50% of these vacuoles exhibited loss of normal vacuolar organisation (Figure 5b) and we saw frequent examples of asynchronous replication, with only single parasites within a vacuole containing daughter cell scaffolds (Figure 5a, inset).

To examine the effects of loss of ZFT in more detail, we imaged parasites by TEM. Although there were no obvious morphological changes, at 5 days post knockdown we observed a heavily vesiculated parasitophorous vacuolar membrane (PVM) with the accumulation of small vesicles in the intravacuolar space. In contrast, the PVM of untreated parasites appeared smooth, with no

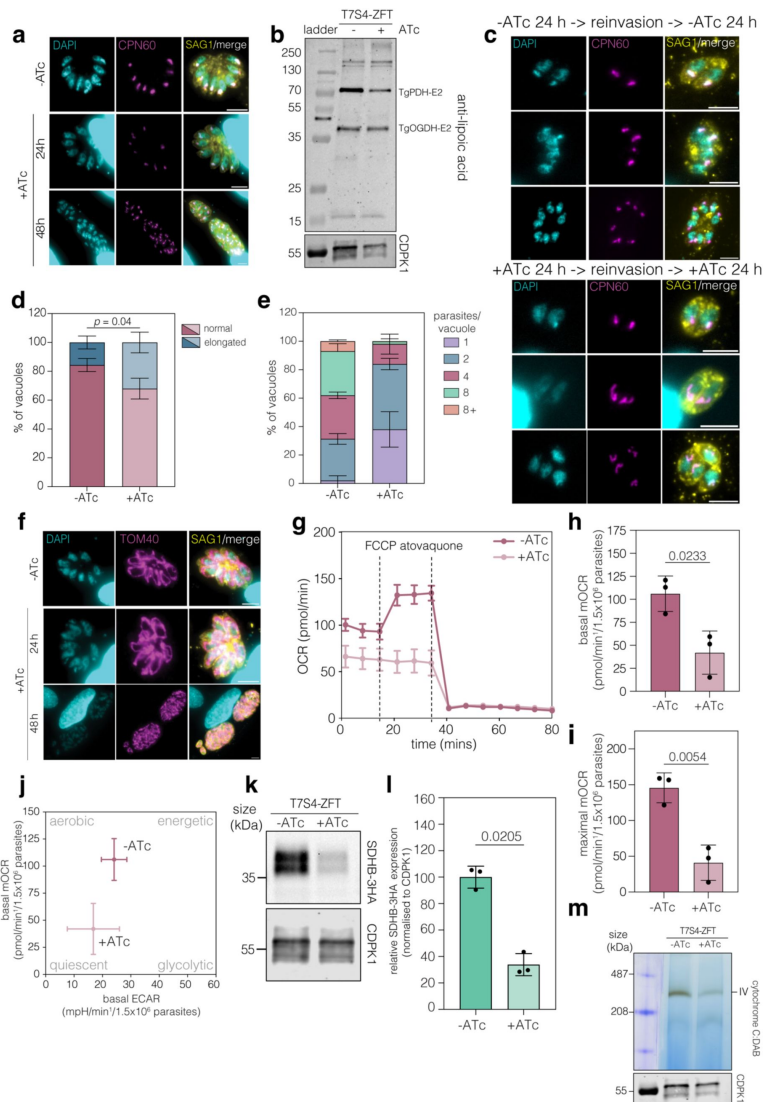


Figure 4.

ZFT knockdown reduces mitochondrial respiration and the expression of Fe-S protein SDHB.

a. Immunofluorescence assay of T7S4-ZFT-3HA_{sag1} in the absence and presence of ATc for 24 h and 48 h, staining for CPN60 (apicoplast marker). **b.** Western blot analysis of T7S4-ZFT with and without ATc 48 h staining for lipoic acid. TgPDH-E2 and TgOGDH-E2 were detected 48 h post ZFT knockdown, suggesting the apicoplast is functional. **c.** Immunofluorescence assay of T7S4-ZFT parasites with and without ATc staining for CPN60 (apicoplast marker) following a further round of mechanical release and invasion to examine if ZFT knockdown has a delayed death phenotype on the apicoplast. Scale bar 5 μ m. **d.** Quantification of % of apicoplasts from (c) with and without ATc that appeared normal or elongated. 100 vacuoles for each condition were quantified. Bars at mean of three independent replicates, $\pm$ SD. **e.** Quantification of the percentage of parasites per vacuole with and without ATc following a further round of mechanical release and invasion. 100 vacuoles for each condition were counted. Bars at mean of three independent replicates, $\pm$ SD. **f.** Immunofluorescence assay as above, staining for TOM40 (mitochondrial marker). Scale bar 5 μ m. **g.** Mitochondrial oxygen consumption rate (mOCR) of T7S4-ZFT, untreated or ATc treated for 48 h via Seahorse assay. Quantification of basal mOCR (**h**) and maximal mOCR (**i**). Bars at mean of three independent experiments, $\pm$ SD. *p* values are from two-tailed unpaired *t* test with Welch's correction. **j.** Metabolic map demonstrating ZFT knockdown shifts parasites to a more quiescent state. Points at mean, $\pm$ SD. **k.** Western blot of T7S4-ZFT SDHB-3HA, untreated and ATc treated for 48 h. CDPK1 used as a loading control. **l.** Quantification of SDHB-3HA expression, relative to uninduced, normalised to CDPK1. Bars at mean of three independent experiments, $\pm$ SD. *p* values are two-tailed paired *t* test. **m.** Complex IV activity assay of T7S4-ZFT parasites with and without ATc 48 h showing ZFT knockdown reduces complex IV activity. CDPK1 used as a loading control.

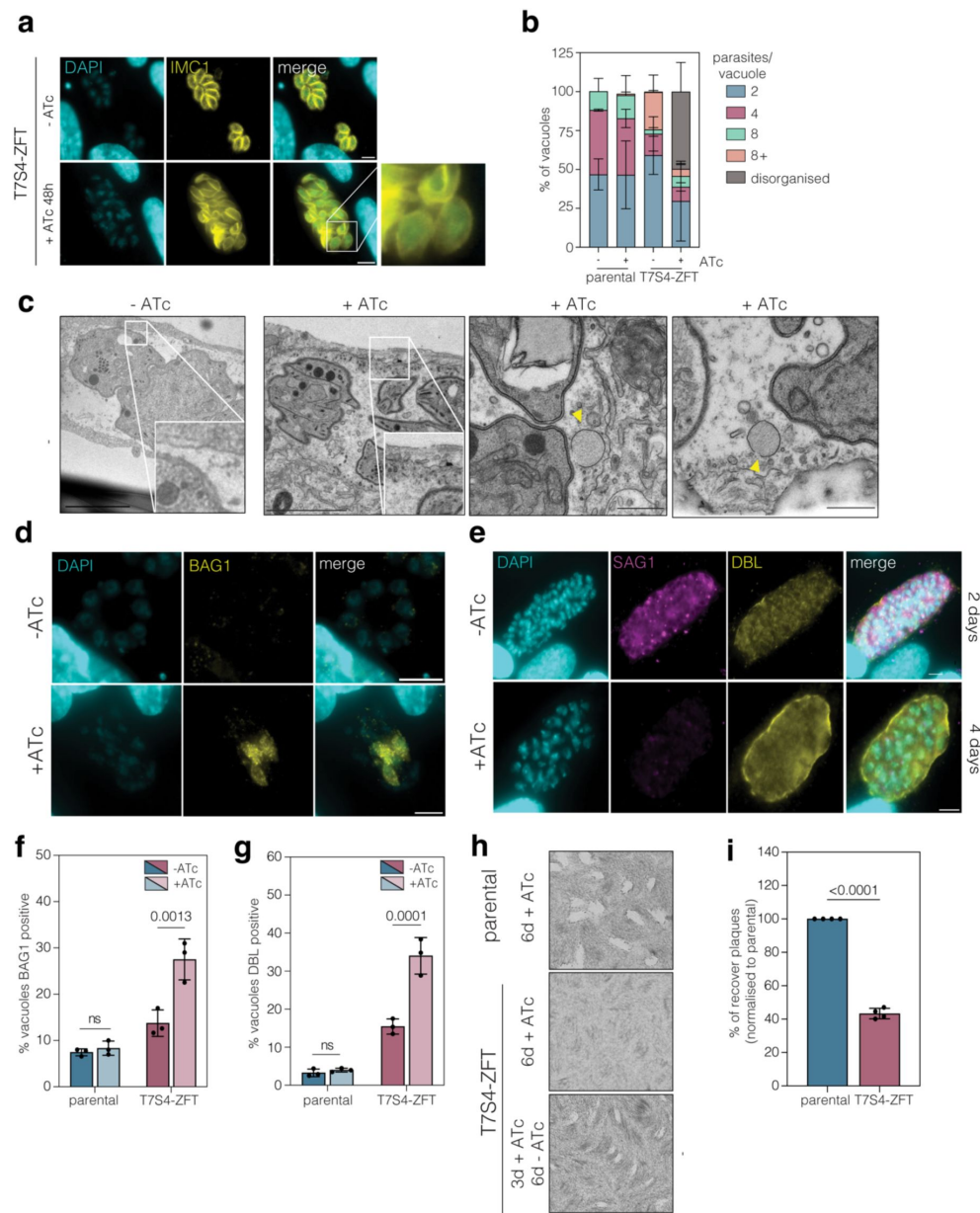


Figure 5.

Knocking down ZFT triggers partial bradyzoite differentiation.

a. Immunofluorescence of T7S4-ZFT parasites uninduced or induced with ATc for 48 h. Highlighted is a region where parasites are disorganised and asynchronously replicating. Scale bar 5 μ m. **b.** Quantification of the percentage of parasites/vacuole in parental and T7S4-ZFT parasites in the presence and absence of ATc for 48 h. 100 vacuoles for each condition were counted. Bars at mean of four independent replicates, $\pm$ SD. **c.** TEM images of T7S4-ZFT parasites with and without ATc for five days. Detail highlights regions of PVM. Yellow arrows point to enlarged vesicles in the PV space. Scale bar 500 nm. **d.** Immunofluorescence of T7S4-ZFT-3HA_{Sag1} untreated and + ATc 72 h, stained with BAG1, a bradyzoite specific marker. Scale bar 5 μ m. **e.** Immunofluorescence assay of T7S4-ZFT-3HA_{Sag1} untreated and + ATc 4 days, stained with SAG1 and DBL, a cyst wall-binding lectin. Scale bar 5 μ m. **f.** Quantification of % vacuoles expressing BAG1 (f) and DBL (g). 100 vacuoles for each condition counted, bars at mean of three independent experiments, $\pm$ SD. *p* values from two-way ANOVA, Tukey corrected for multiple comparisons. **h.** Plaque assays of parental line treated with ATc for 6 days, and T7S4-ZFT parasites, treated either with ATc for 6 days, or for 3 days following washout and 6 days recovery. **i.** Quantification of plaque number recovery following ATc washout, normalised to parental parasites. Bars at mean of four independent experiments, $\pm$ SD. *p* values from two-tailed unpaired *t* test.

vesicular accumulation (**Figure 5c**). Further, we observed multiple enlarged, electron-lucent vesicles upon ZFT knockdown which were not presented in the untreated cells (**Figure 5c**). Similar large vesicles have previously been reported to contain materials that form the cyst wall (Tomita et al., 2013), a characteristic of *T. gondii* conversion to bradyzoites. Previously, we and others have demonstrated that loss of iron is sufficient to drive bradyzoite conversion (Renaud et al., 2024; Sloan et al., 2025; Ying et al., 2024). To determine if iron depletion could mimic this phenotype, parental parasites were treated with DFO for 5 days. As with the knockdown of ZFT, iron-depleted parasites also exhibit similar large vesicles close to the PV membrane (**Figure S4j**).

To determine if loss of ZFT is sufficient to induce stage conversion, we examined the expression of bradyzoite specific markers upon ZFT knockdown. BAG1 is bradyzoite specific protein (Kannan et al., 2019) and DBL is a lectin which binds to the heavily glycosylated cyst wall (Tomita et al., 2013). At 72 and 96 h post ATc addition we saw vacuoles expressing BAG1 (**Figure 5d**) and with peripheral DBL staining around the vacuole (**Figure 5e**). We quantified this and saw a significantly higher percentage of vacuoles expressing BAG1 and DBL ($p = 0.0013$, $p = 0.0001$ respectively, two-way ANOVA, Tukey corrected for multiple comparisons) upon ZFT knockdown (**Figures 5f, g**). These data demonstrate ZFT depletion leads to the expression of the bradyzoite-specific markers BAG1 and DBL.

T. gondii differentiation allows cells to survive otherwise fatal stresses, and parasites can reactivate upon removal of the stress (Cerutti et al., 2020; Soete et al., 1993; Sullivan and Jeffers, 2012). To determine if parasites initiating conversion upon ZFT knockdown remained viable, we quantified parasite survival. Parasites were treated with ATc for 72 h, washed, then incubated without ATc for a further 6 days. After ATc washout, plaques formed, indicating that some parasites remained viable (**Figure 5h**). Upon quantification, around 40 % of parasites were able to reestablish growth after 72 h ZFT knockdown ($p < 0.0001$, two-tailed unpaired *t* test) (**Figure 5i**), a similar proportion to those expressing the bradyzoite marker BAG1 (**Figure 5f**). These results imply that although many Rh parasites are unable to survive a transient loss of ZFT, around 40% of vacuoles initiate transition to bradyzoites and can survive transient loss of ZFT.

ZFT knockdown cannot be rescued by excess zinc

As ZIP-domain proteins commonly transport zinc, we next aimed to assess if ZFT also plays a role in zinc uptake in *T. gondii*. Similar to addition of excess iron, excess zinc was also unable to rescue plaque number (**Figures 6a, b**) or area (**Figure 6c**) in the absence of ZFT. As with iron, changing the promoter alone significantly ($p < 0.0001$, ordinary one-way ANOVA, Tukey corrected for multiple comparisons) sensitised the parasites to excess zinc, leading to a decrease in plaque size (**Figure 6d**). We also tested a zinc ionophore (2-Mercaptopyridine N-oxide), however the zinc ionophore was also unable to rescue the growth phenotype upon knockdown of ZFT (**Figures S5a, b**). As we had previously examined the role of iron, we next tested if both excess iron (200 μ M FAC) and zinc (25 μ M ZnSO₄) were required to rescue the growth phenotype upon ZFT knockdown (**Figure 6a**). However, addition of both metals was also unable to rescue the T7S4-ZFT growth defect (**Figures 6a, b**). Interestingly, changing the promoter alone leads a significant ($p < 0.0001$, two-way ANOVA, Tukey corrected for multiple comparisons) decrease in plaque numbers (to around ~20%) compared to parental parasites in the iron and zinc conditions (**Figure S5c**), suggesting that overexpressing ZFT makes the parasites hypersensitive to both iron and zinc. We also tested the sensitivity of the ZFT-Ty overexpression line to zinc. As suggested by the plaque assays, overexpression of ZFT lead to a small but significant ($p = 0.01$, two-tailed unpaired *t* test) increase in zinc sensitivity (**Figures 6e**, **S5d**).

We next aimed to understand if the expression of ZFT is dependent on zinc availability. We treated ZFT-3HA_{ZFT} parasites with either 50 μ M ZnSO₄ or 5 μ M TPEN, a cation chelator frequently used to reduce zinc levels (Schaefer-Ramadan et al., 2019) for 18 h and 24 h. Unlike for iron, excess zinc did not alter ZFT-3HA_{ZFT} expression at 18 h (**Figures 6f, g**) or 24 h post treatment (**Figures 6h**,

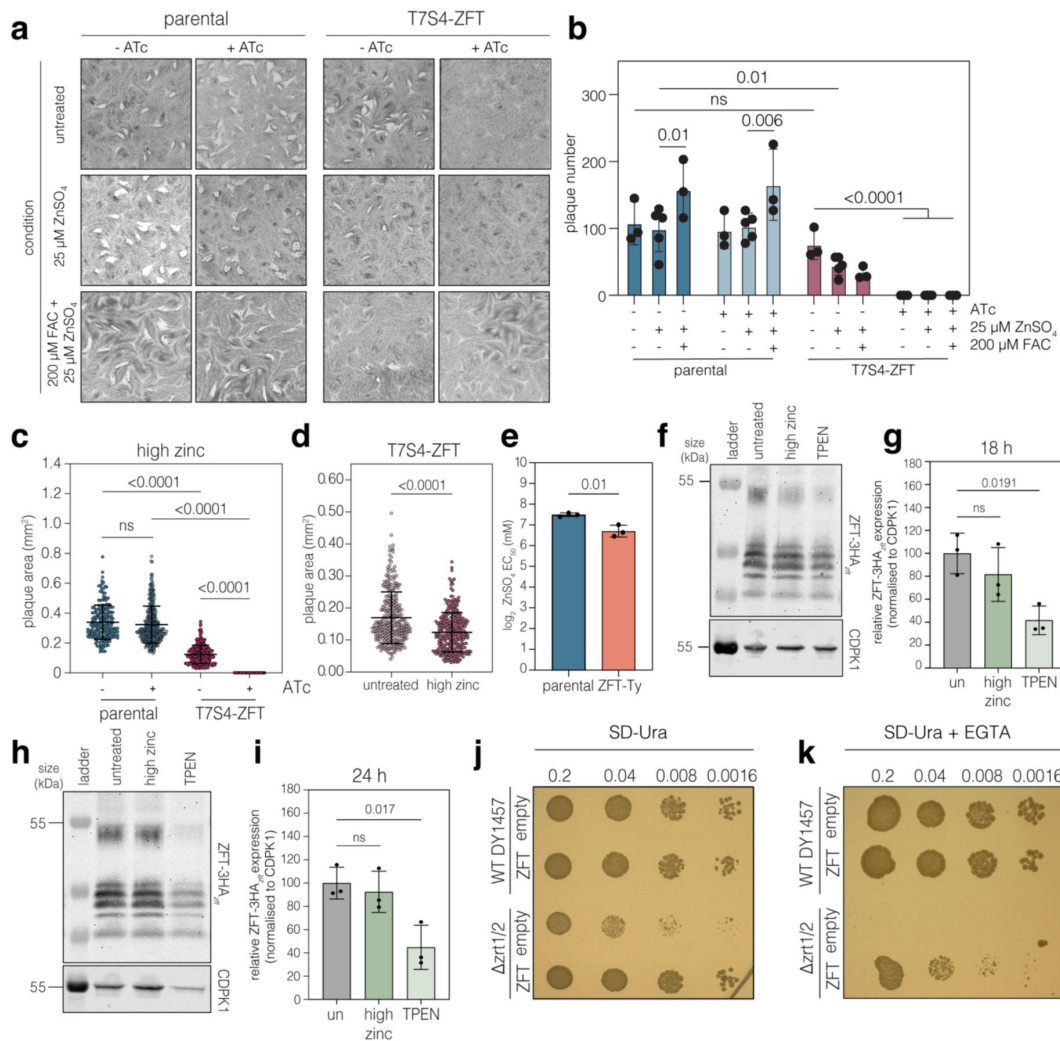


Figure 6.

ZFT expression depends on the availability of zinc, and ZFT expression complements lack of zinc transport in *S. cerevisiae*.

a. Plaque assays showing that the addition of 25 μM ZnSO_4 , and the combination of 25 μM ZnSO_4 and 200 μM FAC, cannot rescue ZFT knockdown. **b.** Number of plaques after treatments. Bars at mean of $n=3$, $\pm$ SD. p values from two-way ANOVA, Tukey corrected for multiple comparisons. **c.** Plaque areas of parasites treated with excess zinc (25 μM ZnSO_4). Points represent individual plaque areas from two (parental -ATc) or three (all other conditions) independent experiments. Bars at mean $\pm$ SD. p values from ordinary one-way ANOVA, Tukey corrected for multiple comparisons. **d.** Plaque area from T7S4-ZFT parasites in untreated and high zinc (25 μM ZnSO_4) conditions. Points represent individual plaque areas from three independent experiments. Bars at mean $\pm$ SD. p values are from two-tailed unpaired t test with Welch's correction. **e.** Graph showing mean EC_{50} for zinc for parental and ZFT-Ty parasites. Bars at mean of $n=3$, $\pm$ SD. p values from two-tailed unpaired t test. **f.** Western blot analysis showing ZFT-3HA_{ZFT} expression levels 18 h post infection in untreated, high zinc (50 μM ZnSO_4) and 5 μM TPEN conditions. CDPK1 used as a loading control. **g.** Quantification of ZFT-3HA_{ZFT} levels at 18 h from three independent experiments. Bars at mean $\pm$ SD. p values from one-way ANOVA, Tukey corrected for multiple comparisons. **h.** Western blot as above, at 24 h post infection. **i.** Quantification of ZFT-3HA_{ZFT} levels at 24 h from three independent experiments. Bars at mean $\pm$ SD. p values from one-way ANOVA, Tukey corrected for multiple comparisons. **j.** Spot assay showing TgZFT expression rescues growth in $\Delta\text{zrt1/2}$ *S. cerevisiae*. Representative of two independent biological experiments. **k.** Under zinc chelation (5 mM EGTA), TgZFT allows $\Delta\text{zrt1/2}$ growth. Representative of two independent biological experiments.

i [↗](#)), and we saw no change in localisation by immunofluorescence (**Figure S5e** [↗](#)). However, at both timepoints, TPEN treatment led to a significant ($p = 0.0191$ and $p = 0.017$ respectively, one-way ANOVA, Tukey corrected for multiple comparisons) reduction in ZFT (**Figures 6f** [↗](#), **g**, **h**, **i**).

ZFT expression complements lack of zinc transport in *S. cerevisiae*

To investigate whether ZFT has a functional role in zinc uptake in *T. gondii*, we first codon optimized the full length ZFT sequence for expression in *Saccharomyces cerevisiae*. This was then constitutively expressed in either the parental strain (DY1457) or a strain lacking both the low and high affinity zinc transporters, $\Delta zrt1/2$ (Ullah et al., 2018 [↗](#)). Under standard conditions, expression of TgZFT complemented the mild growth defect of the $\Delta zrt1/2$ mutant (**Figure 6j** [↗](#)). Upon chelation of zinc by EGTA, the $\Delta zrt1/2$ mutant is unable to grow, however this is rescued by expression of TgZFT (**Figure 6k** [↗](#)). These data demonstrate ZFT expression complements the loss of zinc transport in the $\Delta zrt1/2$ line, confirming ZFT as a functional zinc transporter. Based on this success, we also attempted to complement an iron uptake ($\Delta fet2/3$) yeast mutant. However, despite multiple attempts, we were unable to express codon-optimised ZFT in the $\Delta fet2/3$ yeast mutant. The reason for this was not clear, but it is possible that the $\Delta fet2/3$ yeast strain is less able to survive transformation, or that in this background, expression of ZFT is toxic.

Knocking down ZFT results in a decrease in parasite associated metal

To determine if loss of ZFT led to changes in the metal content of parasites, we used the fluorescence iron indicator FerroOrange, which increases fluorescence in the presence of labile iron (Weber et al., 2020 [↗](#); Yu et al., 2022 [↗](#)). To validate its use, we pre-treated parasites with 200 μ M FAC intracellularly for 48 h before live imaging (**Figure 7a** [↗](#)). Parasites preloaded with iron have a significantly ($p < 0.0001$, two-tailed unpaired t test with Welch's correction) higher MFI of FerroOrange compared to untreated (**Figure 7a** [↗](#)), validating the sensitivity of this indicator. Upon 48 h KD of ZFT, the MFI of FerroOrange was significantly ($p = 0.0027$, two-tailed unpaired t test with Welch's correction) reduced, compared to untreated parasites (**Figure 7b** [↗](#)). We also used the FluoZin-3 indicator which increases in fluorescence in the presence of zinc (Li et al., 2009 [↗](#); Rajapakse et al., 2017 [↗](#)). As above, to validate we saw that zinc-treated parasites had significantly ($p < 0.0001$, two-tailed unpaired t test with Welch's correction) higher fluorescence than untreated cells (**Figure 7c** [↗](#)). However, upon 24 h KD of ZFT, the MFI of FluoZin-3AM was significantly ($p < 0.0001$, two-tailed unpaired t test with Welch's correction) reduced compared to untreated parasites (**Figure 7d** [↗](#)), suggesting lower parasite zinc levels in the absence of ZFT.

To further examine the effect of knocking down ZFT on parasite-associated metal levels, we performed inductively coupled plasma-mass spectrometry (ICP-MS), an elemental analysis technique used to quantify elemental abundance from complex environments (Al-sandaqchi et al., 2018 [↗](#); Hare et al., 2012 [↗](#)). At both 24 h and 48 h ATc treatment, we saw no significant changes in the amount of iron in the parasites (**Figure 7e** [↗](#)). However, iron levels were very close to the limit of detection, even in untreated cells, and iron detection by ICP-MS can be challenging due to interference (Segura et al., 2003 [↗](#)). In contrast, zinc levels were notably higher, and at both 24 h and 48 h ATc treatment, ZFT knockdown parasites had significantly ($p = 0.0062$, and $p = 0.0073$, two-tailed paired t test) reduced parasite associated zinc (**Figure 7f** [↗](#)).

Iron can also be quantified at subcellular resolution using X-ray fluorescence microscopy (XFM) (Aghabi et al., 2023 [↗](#); Miller and Ralle, 2024 [↗](#); Paunesku et al., 2021 [↗](#)), which allows both spatial and quantitative data. Using XFM we were able to detect iron in extracellular parasites (**Figure 7g** [↗](#)), which frequently appeared localised to distinct foci in the parasites, which we believe to be the PLVAC, based on our previous observations (Aghabi et al., 2023 [↗](#)). However, by 48 h post ATc induction we were no longer able to detect iron within the parasites. We also assessed zinc localisation. In untreated cells, zinc localised to what we believe is the parasite nucleus (**Figure 7h** [↗](#)), as has been seen in other organisms (Olea-Flores et al., 2022 [↗](#)).

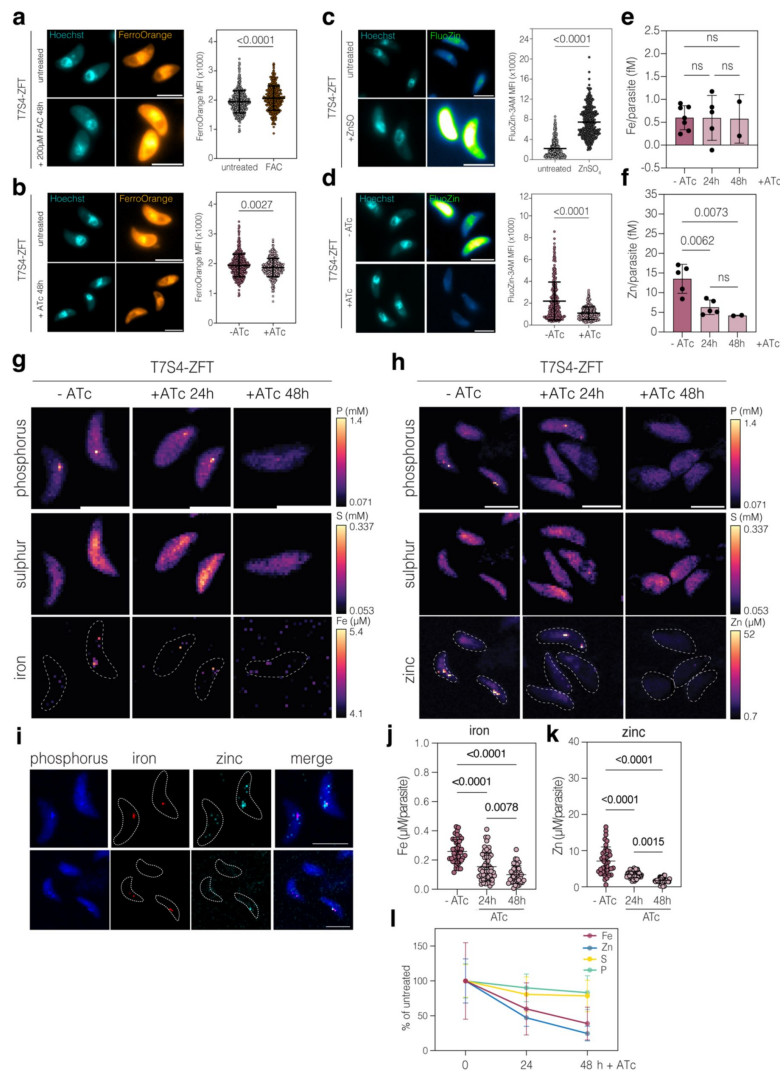


Figure 7.

Knocking down ZFT results in a decrease in parasite associated metal.

a. Live cell imaging and quantification of FerroOrange stained parental parasites, untreated or treated with 200 μ M FAC for 48 h. Scale bar 5 μ m. Points represent the MFI of FerroOrange/parasite from four independent experiments. Bars at mean $\pm$ SD. p value from two-tailed unpaired t test with Welch's correction. **b.** Live cell imaging and quantification of FerroOrange-stained T7S4-ZFT parasites with and without ATc for 48 h. Points are MFI of FerroOrange/parasite from four independent experiments. Bars at mean $\pm$ SD. p value from two-tailed unpaired t test with Welch's correction. **c.** Live cell imaging and quantification of FluoZin-3AM stained parental parasites treated with 50 μ M ZnSO_4 for 24 h. Points represent MFI of FluoZin-3AM/parasite from three independent experiments. Bars at mean $\pm$ SD. p value from two-tailed unpaired t test with Welch's correction. **d.** Live cell imaging and quantification of FluoZin-3AM stained T7S4-ZFT parasites with and without ATc for 48 h. Scale bar 5 μ m. Points represent the MFI of FluoZin-3AM per parasite from three independent experiments. Bars at mean $\pm$ SD. p value from two-tailed unpaired t test with Welch's correction. **e.** ICP-MS quantification of iron/parasite (fM) from T7S4-ZFT parasites with and without ATc at 24 h and 48 h. Bars at mean ($n = 7$ for - ATc, $n = 5$ for + ATc 24 h and $n = 2$ for + ATc 48 h) $\pm$ SD. p values from one-way ANOVA, Tukey corrected for multiple comparisons. **f.** ICP-MS quantification of zinc/parasite (fM) from T7S4-ZFT parasites with and without ATc 24 h and 48 h. Bars at mean ($n = 5$ for - ATc and + ATc 24 h and $n = 2$ for + ATc 48 h) $\pm$ SD. p values from one-way ANOVA, Tukey corrected for multiple comparisons. **g.** XFM of extracellular T7S4-ZFT parasites, untreated and treated for 24 and 48 h, showing phosphorus, sulphur, iron (g) and zinc (h). **i.** XFM of untreated parasites showing iron and zinc colocalisation, along with phosphorus. Quantification of Fe (j) and Zn (k) following ZFT knockdown. Bars at mean $\pm$ SD. p values from one-way ANOVA, Tukey corrected for multiple comparisons. **l.** Quantification of changes in Fe, Zn, S and P as a percentage of untreated cells, following the addition of ATc. Bars at mean $\pm$ SD. Scale bar for all images 5 μ m.

We also saw specific foci of zinc within the cells, which, based on previous results (Chasen et al., 2019), we assume to be the PLVAC. Indeed, we found that iron and zinc, along with phosphorus frequently colocalised in extracellular parasite (**Figure 7i**), as previously has been suggested (Chasen et al., 2019). As with iron, depletion of ZFT led to a decrease in parasite associated zinc, with a loss of zinc both within the nuclei and within foci, which was very evident by 48 h post knockdown (**Figure 7h**).

Parasite associated metals were quantified in untreated parasites and at both 24 and 48 h post ZFT knockdown by XFM. At both timepoints, knockdown parasites have significantly ($p < 0.0001$, ordinary one-way ANOVA, Tukey corrected for multiple comparisons) less parasite-associated iron, from ~250 nM in untreated, to ~150 nM at 24 h and ~95 nM at 48 h (**Figure 7j**). At both 24 h and 48 h KD of ZFT, *T. gondii* parasites have significantly ($p < 0.0001$, ordinary one-way ANOVA, Tukey corrected for multiple comparisons) less parasite associated zinc, from ~6 µM in untreated, to ~3.5 µM at 24 h and ~1.7 µM at 48 h (**Figure 7k**). Interestingly, we found that the concentrations of both sulphur ($p = 0.0014$, ordinary one-way ANOVA, Tukey corrected for multiple comparisons) (**Figure S6a**), and phosphorus ($p = 0.008$, ordinary one-way ANOVA, Tukey corrected for multiple comparisons) (**Figure S6b**), were also significantly reduced at 48 h following ZFT knockdown, although the magnitude of the decrease was much smaller. Comparison of the changes demonstrated a clear, time-dependent depletion of both iron and zinc, with much smaller drops (around 10-20%) in sulphur and phosphorus (**Figure 7l**). Both sulphur and phosphorus are highly abundant in cellular macromolecules, and this small decrease may speak to defects in protein synthesis or DNA replication upon ZFT knockdown.

Together, these data validate the role of ZFT as the major iron and zinc transporter of *T. gondii*.

Discussion

Here we have identified and characterised the first ZIP-domain containing protein in *Toxoplasma*, demonstrating that ZFT is the major iron and zinc importer for the parasite. Sequence analysis of ZFT shows conservation of essential metal coordinating residues and identified a new motif, conserved in coccidia, in the M1 metal binding domain, although with an unknown role in transport. ZFT also encodes a leucine at position 403, previously associated with broad substrate specificity (Gyimesi et al., 2019), which fits with our data that ZFT transports both iron and zinc. However, future studies will examine the substrate preference and kinetics of this transporter in greater detail.

ZIP proteins mediate metal transport both across the plasma membrane, and from vesicles into the cytosol. The localisation of ZFT appears highly dynamic through the parasite's intracellular replication, moving from the plasma membrane to basally clustered intracellular vesicles, after 2-3 rounds of replication. This vacuole stage-dependent localisation is highly unusual in *Toxoplasma* and has not previously been observed. Unfortunately, fluorescent tags led to significant mislocalisation of ZFT to the ER (not shown), so we were unable to observe this trafficking in live cells. Currently, our working model is that ZFT is most active at the plasma membrane, enabling rapid metal incorporation into new vacuoles, and is then internalised and later degraded, presumably to prevent toxic metal overload (Aghabi et al., 2023). This may explain the persistence of plasma-membrane localised ZFT in iron starvation, and its degradation in excess iron. However, from our current results, we cannot exclude the hypothesis that the parasite is taking up extracellular material in digestive vesicles, and ZFT transports metals freed from digested proteins into the cytosol. Future studies examining the kinetics of metal uptake, and the localisation dynamics of ZFT in intracellular parasites in more detail may be able to address this remaining question.

In *Toxoplasma*, vacuolar iron (Aghabi et al., 2023 [↗](#)) and zinc (Chasen et al., 2019 [↗](#)) transporters have been identified with roles in metal storage and detoxification. However, ZFT is the first *Toxoplasma* metal uptake transporter identified and characterised. This transporter is essential, highlighting the essential role of iron and zinc to the parasite. Interestingly, and unlike in *Plasmodium* (Loveridge and Sigala, 2024 [↗](#); Sahu et al., 2014 [↗](#)), we were unable to complement the loss of this transporter by addition of exogenous metals. One possible explanation could be lack of metal access to the PV, however our metal dye experiments and the growth phenotype after treatment increases our confidence that exogenous metals reach the vacuole. A more compelling argument is that ZFT is both essential and sufficient for iron and zinc transport, and in its absence, there are no other transporters able to complement for its loss. As there are three other predicted ZIP transporters in *Toxoplasma*, this raises the questions of their functions within the cell, which will be addressed in future studies.

Knockdown of ZFT triggers differentiation into bradyzoites, which complements previous studies demonstrating a similar phenotype upon iron chelation (Renaud et al., 2024 [↗](#); Ying et al., 2024 [↗](#)). The pathway of iron-starvation triggering differentiation is not yet known, although it is compounded by the action of BFD1 (Sloan et al., 2025 [↗](#)). Interestingly, loss of ZFT leads to a significant inhibition of mitochondrial respiration, with the depletion of the essential, iron-bound protein SDHB (Maclean et al., 2021 [↗](#)) and reduction of complex IV activity. Inhibition of respiration has previously been linked to bradyzoite conversion (Tomavo and Boothroyd, 1995 [↗](#)), however we do not yet know the signalling factors linking iron, zinc or mitochondrial function to bradyzoite differentiation. Iron is known to be required in the apicoplast (Renaud et al., 2022 [↗](#)), although a potential role for zinc in this organelle has not yet been established. We see a delayed phenotype on the apicoplast, suggesting that metal import is also required in this organelle, although no apicoplast metal transporters have yet been identified. While the use of iron within the parasite has been previously examined, the role of zinc in *Toxoplasma* is largely unknown. XFM suggests a concentration of zinc within the nucleus, where it is expected to play a role in DNA replication and gene expression (Olea-Flores et al., 2022 [↗](#)). Beyond the nucleus, zinc is clearly concentrated within one or two foci, in agreement with zinc storage in the PLVAC, as previously hypothesised (Chasen et al., 2019 [↗](#)). Future studies will be needed to unpick both the specific roles of zinc in *Toxoplasma*, and the parasite's response to zinc starvation, as these are likely highly heterogenous.

ZFT appears to be highly regulated by iron availability, at the mRNA (Sloan et al., 2025 [↗](#)), protein, and localisation levels. However, ZFT is not regulated by excess zinc, although zinc-mediated ZIP-protein degradation has been observed in other organisms (e.g. (Pang et al., 2023 [↗](#))). Interestingly, there is evidence from other organisms that excess zinc modulates both iron uptake (Blaby-Haas and Merchant, 2013 [↗](#); Xu et al., 2019 [↗](#)) and copper homeostasis (Xu et al., 2019 [↗](#)). It is possible that excess zinc leads to a decrease in iron acquisition, stabilising ZFT expression, however this has not yet been assessed. Previously, we have shown that ZFT is stabilised under iron starvation (Sloan et al., 2025 [↗](#)), however we did not see the same effect after treatment with the zinc chelator TPEN. There has not yet been any investigation into potential zinc-mediated regulation in *Toxoplasma*, and so we are currently unable to say if this is due to a lack of zinc sensing and response in these parasites, or the treatment or timings we used was insufficient to trigger such a pathway.

The use of XFM to quantify parasite-associated metals has been vital in this work. Beyond localising iron, zinc and phosphorus to an overlapping compartment in extracellular parasites, we were able to show that knockdown of ZFT lead to significant decreases in parasite-associated iron and zinc. There are some limitations of XFM, e.g. we were unable to quantify copper or manganese using this approach, probably due to low concentrations within the parasite. Further, for technical reasons we worked with chemically fixed parasites which may have led to an underestimate of metal concentrations (Jin et al., 2017 [↗](#)). However, this provides direct evidence of changes in the elemental composition of parasites in the absence of ZFT.

In summary, here we characterise the first iron and zinc uptake transporter in *Toxoplasma*. ZFT is indispensable, highlighting the importance of iron and zinc uptake for the parasite. The identification of this transporter opens new avenues for future investigation into the essential metallobiology of these parasites.

Materials and methods

Toxoplasma gondii and host cell maintenance

Toxoplasma gondii tachyzoites were grown in human foreskin fibroblasts (HFFs) cultured in Dulbecco's modified Eagle's medium (DMEM), supplemented with 3% (D3) heat-inactivated foetal bovine serum (FBS), 2 mM L-glutamine and 50 U/mL Penicillin/Streptomycin and maintained at 37 °C with 5% CO₂.

Toxoplasma gondii strain construction

ZFT-3HA_{zft} and ZFT-3HA_{sag1} parasite strains were generated as previously described (Sloan et al., 2025 [\[4\]](#)). To generate RHtdTomato pTUB8 ZFT-Ty-DHFR parasites, ZFT was amplified from cDNA and a Ty tag was inserted at the 3' end of the gene by PCR using primers P1 and P2 (Table 2 [\[4\]](#)). The tubulin promoter TUB8 plasmid was digested with EcoRI and PacI and then assembled with the ZFT-Ty PCR product to generate the new plasmid. The DHFR sequence was amplified using primers P3 and P4. The TUB-ZFT-Ty plasmid was then digested using BbcC1 enzyme then the DHFR sequence was inserted into the plasmid. TUB8-ZFT-Ty-DHFR plasmid was linearised using enzyme ScaI and the transfected into RHtdTomato parasites using a Gene Pulse BioRad electroporator. Parasites were selected using pyrimethamine and cloned by limiting dilution. Clones were confirmed using primers P5 and P6 (Table 2 [\[4\]](#)). For the generation of the F3ΔHX T7S4-ZFT strain, the T7S4 promoter was amplified from the G13m5 plasmid (Sheiner et al., 2011 [\[4\]](#)) using primers P7 and P8. F3ΔHX parasites were transfected using a Gene Pulse BioRad electroporator with 30 µg of the purified repair template T7S4-ZFT PCR product and 40 µg of a plasmid containing Cas9 guide targeting the 3' end of the genomic sequence of ZFT (Table 1 [\[4\]](#)). Parasites were selected with pyrimethamine and cloned by limiting dilution. Clones were screened for 3' integration using primers P9 and P10, and 5' integration using primers P11 and P12. For the generation of the F3ΔHX T7S4-ZFT-3HA_{sag1} line, ZFT was tagged as previously described (Sloan et al., 2025 [\[4\]](#)) in F3ΔHX T7S4-ZFT parasites, selected for with chloramphenicol, and cloned by limiting dilution. The T7S4-ZFT SDHB-3HA and T7S4-ZFT ABCE1-3HA parasite strains were generated by tagging SDHB and ABCE1 as previously described (Maclean et al., 2024 [\[4\]](#), 2021 [\[4\]](#)) in the F3ΔHX T7S4-ZFT parasites, selected for with chloramphenicol, and cloned by limiting dilution. T7S4-ZFT SDHB-3HA clones were screened for 5' integration using primers P13 and P15, and 3' integration using primers P14 and P16. T7S-ZFT ABCE1-3HA clones were screened for 5' integration using primers P13 and P17, and 3' integration using primers P14 and P18. Parasite lines were selected using either chloramphenicol (Sigma Aldrich, C3175), made up in MiliQ water and used at 40 µg/ml or pyrimethamine (Sigma Aldrich, 46706), made up in ethanol and used at 2 µM. All sgRNA sequences (Table 1 [\[4\]](#)) and primers (Table 2 [\[4\]](#)) used in the study are listed below.

Chemicals

Ferric ammonium citrate (FAC) (Sigma Aldrich, F5879) and Deferoxamine mesylate salt (DFO) (Sigma Aldrich, D9533), were made up in 1X phosphate buffered saline (PBS) and used as indicated. Zinc sulphate (ZnSO₄) (Sigma Aldrich, 307491) and 2-Mercaptopyridine N-oxide sodium salt (Sigma Aldrich, H3261), were made up in distilled water and used at indicated concentrations. N,N,N',N'-tetrakis-(2-pyridylmethyl) ethylenediamine (TPEN) (Merck, 616394), was made up in DMSO and used at 5 µM. Anhydrotetracycline hydrochloride (ATc) (abcam, ab145350), made up in ethanol and used at 0.5 µM, unless indicated.

gRNA Sequences	
Name	Sequence (5' – 3')
ZFT 3'	TGAACTGTTGCACAACTAC
ZFT 5'	CCTTGAGACCTTTCCTCCCG
SDHB 3'	CGAGCCGAGGTTACGCAGCG
ABCE1 3'	GCTCGACGACGCTTGAAGCG

Table 1

A list of the gRNA sequences used in this study.

Primer Sequences		
Name	Sequence (5' – 3')	Notes
P1	ctcgttggcatttttcttgATGGCGACGGTCGCAGAC	ZFT amplification from cDNA Fw
P2	tgagcacaacggtgattaatttaatcgagcgggtcctggttcgtgtggacctcA GCGTCTAGAGCCAGATCCAC	ZFT amplification from cDNA and Ty tag Rv
P3	CTCGGTCTTACCCTGGTGGATCTGGCTCTAGACGCaGA GGTCCACACGAACCAGGACCCGCTCGATAGCGGAGCTA CTAACTTCAGC	P2A-DHFR amplification Fw
P4	GTGATTAATTTAATCGAGCGGGTCCTGGTTCGTGTGGAC CgctaGACAGCCATCTCCATCTGGAT	P2A-DHFR amplification Rv
P5	ctcgttggcatttttcttgatggcgacggtcgcagac	TUB8 ZFT-Ty DHFR integration verification Fw
P6	tgagcacaacggtgattaatttaatcgagcgggtcctggttcgtgtggacctca gctctagagccagatccac	TUB8 ZFT-Ty DHFR integration verification Rv
P7	accgaacagggcgacgattttccccttgagaccttctCTTCGCCAGG CTGTAAATCC	ZFT T7S4 Fw
P8	agctggcagcagacatcacatcgctctgcgaccgtcgccatTGGTTGAAG ACAGACGAAAGCAGTTG	ZFT T7S4 Rv
P9	cctgataggttctggacctccctcgc	ZFT T7S4 5' integration Fw
P10	cgtagagcagaaacgcactactaaag	ZFT T7S4 5' integration Rv
P11	gcaaagccagaggggatatatgctcc	ZFT T7S4 3' integration Fw

Table 2

List of primer sequences used in this study

P12	cgaggtcgacgtatcgataagcttacg	ZFT T7S4 3' integration Rv
P13	AAATAGGATGGTACATCCCCC	HA integration test Rv
P14	gaaacacacttcgtgcagc	CAT integration test Fw
P15	CTTTTCTTTTGTGGGTTGTGC	SDHB-3HA Fw
P16	CGACATTTGCATGTACATACC	SDHB-3HA UTR Rv
P17	ATCTCGTCAGCGGCATGAACC	ABCE1-3HA Fw
P18	CATACACGGACACAGATACGC	ABCE1-3HA UTR Rv
P19	aaaaaaaaatataccccagcctcgagATGGCCACCGTCGCTGATG AC	Amplification of ZFT for yeast expression Fw
P20	aagtcaaagctggatccgcctcgagTTAGTCAAGAGGGTCTTGG TTAG	Amplification of ZFT for yeast expression Rv
P21	GGGACGACATGGAGAAAATC	Actin RT-qPCR Fw
P22	AGAAAGAACGGCCTGGATAG	Actin RT-qPCR Rv
P23	CGGACAACCGAATAGTGGCT	ZFT RT-qPCR Fw
P24	CGCCAACGATCATTCTTCG	ZFT RT-qPCR Rv

Table 2 (continued)

Immunofluorescence Assays

Indirect immunofluorescence assays (IFA) were performed on infected HFF monolayers. Intracellular *T. gondii* parasites were fixed with 4% paraformaldehyde (PFA) for 20 minutes at room temperature and washed with 1X PBS. Cells were permeabilized and blocked in blocking buffer (PBS/0.2% Triton X-100/2% Bovine Serum Albumin (BSA)) for 30 minutes at room temperature, or at 4°C overnight. Coverslips were then incubated in a wet chamber with primary antibodies (rat anti-HA 1:500 (Merck 11867423001), rabbit anti-GAP45 (Plattner et al., 2008 [DOI](#)) 1:2000, a kind gift from Dominique Soldati-Favre, rabbit anti-IMC1 (Wichroski et al., 2002 [DOI](#)) 1:2000, rat anti-SAG1 1:500 (Abcam), rabbit anti-BAG1 (Kannan et al., 2019 [DOI](#); Smith et al., 2021 [DOI](#)) 1:500, a gift from Lilach Sheiner, rabbit anti-TOM40 (Van Dooren et al., 2016 [DOI](#)) 1:1000, a kind gift from Dr. Giel Van Dooren (ANU) and rabbit anti-CPN60 (Agrawal et al., 2009 [DOI](#)) 1:1000, a kind gift from Lilach Sheiner, in blocking buffer for 1 hour at room temperature. Coverslips were then washed three times with PBST (0.2% Triton X-100 in PBS). After washing, slides were incubated in a wet chamber with the secondary antibodies (Goat anti-rat AlexaFluor-594 1:1000 (Invitrogen), goat anti-mouse AlexaFluor-488 1:1000 (Invitrogen), goat anti-rabbit AlexaFluor-488 1:1000 (Invitrogen), goat anti-guinea pig-488 1:1000 (Invitrogen) or Dolichos Biflorus-488 (DBL) 1:1000, in blocking buffer for 1 h at room temperature in the dark. Coverslips were then washed with PBST (0.2% Triton X-100 in PBS), and mounted using Fluoromount with 4',6-diamidino-2-phenylindole (DAPI), to stain DNA (Southern Biotech, 0100-20).

FerroOrange Live Cell Imaging

To analyse the levels of iron in the parasites, FerroOrange (Dojindo, F374-10) was used. Parasites were untreated, treated with ATc, or 200 µM FAC for 48 h. Parasites were released, pelleted and resuspended in DMEM supplemented with 1 µM FerroOrange and 10 µM Hoechst 33342 (ThermoFisher Scientific, H3570), put onto poly-l-lysine treated live cell dishes (Cellvis, D29-20-1.5-N) and incubated at 37 °C for 10 minutes before imaging. Images were obtained using an inverted DiM8 microscope (Leica Microsystems) using LAS X Core software and processed using FIJI software. Intensity was quantified using FIJI (Schindelin et al., 2012 [DOI](#)).

FluoZin-3AM Live Cell Imaging

To analyse the levels of zinc in the parasites, a Zn²⁺ selective indicator FluoZin-3AM (ThermoFisher Scientific, F14201) was used. Parasites were untreated, treated with ATc, or 50 µM ZnSO₄ for 24 h. Parasites were released, pelleted and resuspended in DMEM supplemented with 2 µM FluoZin-3 and 10 µM Hoechst 33342 (ThermoFisher Scientific, H3570), put onto poly-l-lysine treated live cell dishes (Cellvis, D29-20-1.5-N) and incubated at 37 °C for 10 minutes before imaging. Images were obtained using an inverted DiM8 microscope (Leica Microsystems) using LAS X Core software and processed using FIJI software. Intensity was quantified using FIJI (Schindelin et al., 2012 [DOI](#)).

RT-qPCR

To validate the T7S4-ZFT parasites, RT-qPCR was performed to confirm successful knockdown of ZFT mRNA levels. HFFs were infected with T7S4-ZFT with and without ATc for 24, 48, 72 and 96 h before being mechanically lysed and filtered through a 3 mm filter to remove any host cell debris. Parasites were then harvested by centrifugation at 2000 x g for 10 minutes, supernatant was removed, pellet was washed once in sterile PBS before spinning down in a microcentrifuge at 7000 x rpm for 10 minutes. Supernatant was removed and pellets were frozen at -20 °C overnight. RNA was then extracted from the pellets using the RNeasy Mini Kit (Qiagen, 74104). 1 µg RNA was then treated with DNase I 1 U/µL (ThermoFisher Scientific, 18068015) at room temperature for 15 minutes before inactivating the DNase with 25 mM EDTA followed by DNase denaturation at 65 °C for 10 minutes. Using 1 mg of DNase I treated RNA, cDNA synthesis was then performed using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, A48571) according to

manufacturer's instructions. RT-qPCR was carried out on the Applied Biosystems 7500 Real Time PCR machine using PowerSYBR green PCR master mix (ThermoFisher Scientific 4367659) and 2 ng of 1:10 diluted cDNAs per reaction. Primers P21 and P22 were used for actin as a housekeeping control gene and primers P23 and P24 were used for ZFT (**Table 2**). The results are from three independent biological experiments, each performed in technical triplicate. Relative fold changes for - ATc vs. + ATc samples were calculated using the Pfaffl method (Pfaffl, 2001). Graphpad Prism 10 was used to perform statistical analysis.

Transmission electron microscopy

Infected cells were fixed with 2.5% glutaraldehyde and 4% paraformaldehyde in 0.1M cacodylate buffer. Following serial washes in 0.1M cacodylate buffer, the material was post-fixed in 1% OsO₄ and 1.25% K₄[Fe(CN)₆] for 1 hour in the dark, and contrasted en bloc with 0.5% aqueous uranyl acetate for 1 hour at room temperature in the dark. The samples were then dehydrated in ascending acetone series and embedded in epoxy resin. Ultra-thin sections (60nm) were collected using a Leica Ultramicrotome UC6 and observed in a Jeol 1200EX transmission electron microscope (Jeol, Japan).

Plaque Assays

200, 500 or 1000 parental or ZFT-Ty/T7S4-ZFT parasites were applied onto confluent monolayers of HFF cells in a 6-well plate in D3 media, supplemented when indicated with 200 μM FAC, 25 μM ZnSO₄, 2 μM zinc ionophore (2-Mercaptopyridine N-oxide sodium salt (Sigma Aldrich, H3261)), or both 200 μM FAC and 25 μM ZnSO₄, in the presence or absence of 0.5 μM ATc, unless stated otherwise. Parasites were allowed to replicate for 7 days undisturbed, washed 3x with PBS, fixed with ice cold 100 % methanol and stained using crystal violet stain (12.5 g crystal violet in 125 ml ethanol, diluted in 500 ml 1% ammonium oxalate) for 2-3 hours at room temperature before washing with distilled H₂O, drying and imaging. For ATc washout experiments, ATc was removed from cells after 3 days. Cells were then washed 2x with PBS and fresh D3 media was added to the cells. Parasites were allowed to replicate for a further 6 days undisturbed, then fixed and stained as described above.

Fluorescent Growth Assays

Growth assays of fluorescent parasites were performed as previously (Aghabi et al., 2023). Briefly, host cells were infected with 1,000 either RHtdTomato (Elati et al., 2023) or RHtdTom TUB8 ZFT-Ty tachyzoites per well and allowed to invade for 2 hours at 37 °C. The media was then removed and supplemented with either FAC, DFO, ZnSO₄ or Ars (at indicated concentrations) and incubated at 37 °C with 5% CO₂ for 4 days. Total parasite fluorescence was recorded at 4 days using a PHERAstar FS microplate reader (BMG LabTech). All experiments were performed in triplicate with at least three independent biological replicates. Uninfected wells served as blanks and fluorescence values were normalised to infected, untreated wells. Dose-response curves and EC₅₀ were calculated using GraphPad Prism 10.

Western blotting

Parasites were mechanically released, filtered and collected by centrifugation at 1500 x g for 10 minutes, and lysed with RIPA lysis buffer (150 mM sodium chloride, 1 % Triton X-100, 0.5 % sodium deoxycholate, 0.1 % sodium dodecyl sulphate (SDS) and 50 mM Tris, pH 8.0) for 30 minutes on ice. Samples were then resuspended with 5X Laemmli buffer (10% SDS, 50% glycerol, 300 mM Tris-HCL pH 6.8 and 0.05% bromophenol blue), boiled at 95 °C for 5 (unless stated otherwise) and separated on a 10% SDS-PAGE gel, initially at 120V for 20 minutes then 175V for 60 minutes. PageRuler Prestained Protein Ladder (ThermoFisher Scientific, 26619) was used as a molecular weight marker. Proteins were then wet transferred to nitrocellulose membrane (0.2 μM, Cytiva, 10600004) in Towbin buffer (0.025 M Tris, 0.192 M glycine, 10 % methanol) for 60 minutes at 250 mA and blocked for at least 1 h at room temperature in blocking buffer (5 % milk in 0.1 %

Tween/PBS). Blots were then stained with primary antibodies overnight at 4°C (rat anti-HA, Merck 11867423001 1:500, mouse anti-Ty (Bastin et al., 1996 [DOI](#)) 1:1000, rabbit anti-CPN60 (Agrawal et al., 2009 [DOI](#)) 1:2000, a kind gift from Lilach Sheiner, rabbit anti-lipoic acid, Abcam ab58724, 1:1000 and guinea pig anti-CDPK1 (Waldman et al., 2020 [DOI](#)) 1:10,000, both kind gifts from Sebastian Lourido), followed by secondary fluorescent antibodies at room temperature for 1 h (rabbit anti-rat HRP 1:5000 (abcam, ab6845), goat anti-mouse HRP 1:10,000 (Promega, W4021), goat anti-mouse IRDye 680 1:10,000 (Li-Cor) and goat anti-guinea pig IRDye 680 1:10,000 (Li-Cor)). ZFT-3HA_{zft} was detected using Pierce ECL Western Blotting Substrate (ThermoFisher Scientific) and imaged using the Invitrogen iBright 1500 Imaging System. CDPK1 was detected using the LI-COR Odyssey LCX system.

For Blue native (BN) PAGE, 10⁷ parasites were mechanically released, filtered and harvested by centrifugation at 1500 x g for 10 minutes. The parasite pellet was then resuspended in a BN lysis buffer (1% DDM, 750 mM aminocaproic acid, 0.5 mM EDTA and 50 mM Bis-Tris-HCL pH 7.0), and incubated on ice for 30 minutes before being centrifuged at 16,000 x g for 30 minutes at 4 °C. The supernatant containing the solubilized membrane proteins was then added to a sample buffer containing coomassie G250 (NativePAGE), with the final concentration being 0.0625% Coomassie G250 and 0.25% DDM. 10⁷ parasites were loaded onto a 3-12% Native PAGE Bis-Tris gel (ThermoFisher Scientific, NP0301PK2). NativeMark™ was used as the molecular marker. Proteins were then transferred onto a PVDF membrane (0.45 µM, Cytiva, 10600023) via wet transfer in Towbin buffer for 60 minutes at 100 V. Immunoblotting and chemiluminescence detection was performed as above.

Complex IV in gel oxidation assay

Intracellular *T. gondii* tachyzoites were harvested by centrifugation at 1500 x g for 10 minutes at 4 °C. The parasites were washed once with 1x PBS and centrifuged at 12,000 x g for 1 minute at 4 °C. Parasite pellets were then resuspended in BN-PAGE lysis buffer (25% (w/v) 4x Native PAGE buffer (Invitrogen, BN20032), 2% (w/v) protease inhibitors, 1% (v/v) digitonin (Sigma-Aldrich, D141) in distilled water) to a final concentration of 2.5×10⁵ parasites/µL. Samples were kept on ice for 30 minutes then spun at 20,000 x g for 30 minutes at 4 °C. The supernatant was transferred to a fresh labelled tube. 15 µL of sample per well was aliquoted, and ¼ of G250 additive (5% stock) as the detergent % in the sample was added. Samples were run on a 4-16% BN-PAGE gel (Novex-Life technologies) in the cold room at 4 °C with the dark cathode buffer in the centre of the rank at 150V for 60 minutes. The dark cathode buffer was then replaced with the light cathode buffer and run for a further 90 minutes at 250V until the dye has reached the bottom of the gel. To perform the in gel enzyme stain, the oxidation buffer (50 mM OF KH₂PO₄ pH 7.2 in distilled water) was prepared. 1 mg/mL cytochrome c (Sigma-Aldrich, C7752) and 0.1% (w/v) 3'-3'-diaminobenzidine tetrahydrochloride (DAB) (Sigma-Aldrich, D5637) were added to the oxidation buffer. The gel was then gently removed from the plastic cassette and incubated in the staining solution overnight before imaging.

Seahorse Assay / Extracellular flux analysis

The *Toxoplasma* Seahorse assay was performed as previously described (Hayward et al., 2022 [DOI](#)). Briefly, parasites were incubated in the absence or presence of ATc for 48 h, extracellular parasites were removed by washing 1X PBS, and intracellular parasites were mechanically released, filtered and harvested by spinning at 2000 x g for 10 minutes. Parasites were then resuspended in Seahorse XF media (Agilent) supplemented with 5 mM glucose and 1 mM glutamine at 37 °C. 1.5 x 10⁶ parasites in a final volume of 175 µL were added to each well of a poly-L-lysine-treated seahorse XFp cell culture miniplate (Agilent). Parasites were then centrifuged at 300 x g for 5 minutes to ensure sedimentation before the basal oxygen consumption rate (OCR) and the extracellular acidification rate (ECAR) was quantified using a Seahorse XF HS Mini Analyser (Agilent Technologies). The OCR following the addition of 1 µM atovaquone (Scientific, 17575545) was considered as the non-mitochondrial oxygen consumption rate and subtracted as background. The

basal and maximal oxygen consumption rates of the parasites were calculated by subtracting the background from the raw OCR values before and after the addition of 1 μM Carbonyl cyanide 4-(trifluoromethoxy) phenylhydrazone (FCCP) (Merck, C2920) respectively. Basal ECAR values were taken as raw values. This experiment was performed in three independent biological replicates, each with three technical triplicates.

ICP-MS

Total parasite associated iron and zinc were quantified using ICP-MS. Confluent HFFs cells in T150 flasks were infected with T7S4-ZFT parasites for 24 h before the addition of ATc for a further 24 h, or infected with T7S4-ZFT parasites + ATc for 48 h to ensure efficient knockdown. Untreated (-ATc) parasites were used as a control. Parasites were harvested by scraping, syringing and filtering through either 0.3 μm filters or PD-10 columns (Cytiva, 17085101), and quantified using a haemocytometer. Parasites were then washed three times with PBS supplemented with 6% Chelex 100 sodium form (Merck, C7901), then once with 1 mM EDTA before being digested in 65% nitric acid (Merck, 225711) at 85°C for 5 h. Digested samples were then diluted 1:10 in 2% nitric acid with 0.025% Triton X-100. At least three biological replicates were collected, with three technical replicates per biological sample. Samples were quantified at the Scottish Universities Environmental Research Centre (SUREC) and processed by Dr Valerie Olive. Briefly, samples were run on an Agilent 7500ce ICP-MS, fitted with a double pass spray chamber and a self-aspirer nebulizer tuned for 0.1 ml/min. A 3 ppb Indium (^{115}In) was introduced as internal standard to correct for the sensitivity drop caused by the sample matrix. Cu and Zn were taken on the masses 63 and 66 respectively and the results are an average of 10 repeats. The calibration equation was calculated from 3 points at 5, 10 and 20 ppb using Cu and Zn mono-element solutions Cu and Zn specpure®. Iron was taken at the mass 56 using a collision cell with a flow of Hydrogen at 4 ml/min to suppress the interference of the $^{40}\text{Ar}^{16}\text{O}$. The calibration equation was calculated using a 200 ppb and 300 ppb solution of mono-element Fe specpure®, results are an average of 20 repeats.

X- ray Fluorescence Microscopy (XFM)

T7S4-ZFT parasites were treated in the presence or absence of ATc for 24 h or 48 h. Parasites from a fully lysed dish were filtered, pelleted and resuspended in 5 mL of extracellular buffer (116 mM NaCl, 5.4 mM KCL, 0.8 mM MgSO_4 , 5.5 mM glucose, 50 mM HEPES, pH 7.2). Approximately 500 μL of resuspended parasites were then added onto each poly-l-lysine 2 x 2 mm silicon nitride (SiN) window (Norcada, X5200D) and spun down gently at 200 x g for 2 minutes then left at room temperature for 20 minutes.

Following adhesion/infection, cells were then washed twice with 1X chelexed PBS then fixed in 4% PFA made up in 1X chelexed PBS for 20 minutes at room temperature. Cells were then washed twice in 1X chelexed PBS, twice in 100 mM ammonium acetate and finally twice in MiliQ H_2O before drying overnight. Elemental mapping was performed at the European Synchrotron Radiation Facility in Grenoble, France at beamline ID21 under proposal number LS-3419. Data is available from <https://doi.esrf.fr/10.15151/ESRF-DC-2119497225>. Analysis was performed using PyMca5.9.2 (Solé et al., 2007) and FIJI (Schindelin et al., 2012). The 48 h time point is representative of two independent biological experiments, whereas the 24 h time point is representative of one independent biological experiment. We would like to thank Hiram Castillo Michel for assistance and support.

Yeast strain construction

Full length ZFT was cloned into the yeast expression vector pDR195 (AddGene, #36028) using primers P19 and P20 (Table 2). Wildtype (DY1457) and $\Delta\text{zrt1/2}$ (Ullah et al., 2018) strains, a kind gift from Jim Dunwell (University of Reading), were transformed as previously described

(Gietz and Schiestl, 2007 [↗](#)). Positive transformations were selected on synthetic dropout agar lacking uracil (SD-Ura). Colonies were grown overnight in SD-Ura media and spotted onto SD-Ura agar, or SD-Ura agar supplemented with 5 mM EGTA, and incubated for 4 days.

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Additional files

Supplemental Figures [↗](#)

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Reviewer #1 (Public review):

In this manuscript, Aghabi et al. present a comprehensive characterization of ZFT, a metal transporter located at the plasma membrane of the eukaryotic parasite *Toxoplasma gondii*. The authors provide convincing evidence that ZFT plays a crucial role in parasite fitness, as demonstrated by the generation of a conditional knockdown mutant cell line, which exhibits a marked impact on mitochondrial respiration, a process dependent on several iron-containing proteins. Consistent with previous reports, the authors also show that disruption of mitochondrial metabolism leads to conversion into the persistent bradyzoite stage. The study then employed advanced techniques, such as inductively coupled plasma-mass spectrometry (ICP-MS) and X-ray fluorescence microscopy (XFM), to demonstrate that ZFT depletion results in reduced parasite-associated metals, particularly iron and zinc. Additionally, the authors show that ZFT expression is modulated by the availability of these metals, although defects in the transporter could not be compensated for by exogenous addition of iron or zinc.

While the manuscript does not directly investigate the transport function of ZFT through biochemical assays, the authors indirectly support the notion that ZFT can transport zinc by demonstrating its ability to compensate for a lack of zinc transport in a yeast heterologous system. Furthermore, phenotypic analyses suggest defects in iron availability, particularly with regard to Fe-S mitochondrial proteins and mitochondrial function. Overall, the manuscript provides a solid, well-rounded argument for ZFT's role in metal transport, using a combination of complementary approaches. Although direct biochemical evidence for the transporter's substrate specificity and transport activity is lacking, the converging evidence, including changes in metal concentrations upon ZFT depletion, yeast complementation data, and phenotypic changes linked to iron deficiency, presents a convincing case. Some aspects of the results may appear somewhat unbalanced, particularly since iron transport could not be confirmed through heterologous complementation, while zinc-related phenotypes in the parasites have not been thoroughly explored (which is challenging given the limited number of zinc-dependent proteins characterized in *Toxoplasma*). Nevertheless, given that metal acquisition remains largely uncharacterized in *Toxoplasma*, this manuscript provides an important first step in identifying a metal transporter in these parasites, and the data presented are generally convincing and insightful.

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Reviewer #2 (Public review):

Summary:

The intracellular pathogen *Toxoplasma gondii* scavenges metal ions such as iron and zinc to support its replication; however, mechanistic studies of iron and zinc uptake are limited. This study investigates the function of a putative iron and zinc transporter, ZFT. In this paper, the authors provide evidence that ZFT mediates iron and zinc uptake by examining the regulation of ZFT expression by iron and zinc levels, the impact of altered ZFT expression on iron sensitivity, and the effects of ZFT depletion on intracellular iron and zinc levels in the parasite. The effects of ZFT depletion on parasite growth are also investigated, showing the importance of ZFT function for the parasite.

Strengths:

A key strength of the study is the use of multiple complementary approaches to demonstrate that ZFT is involved in iron and zinc uptake. Additionally, the authors build on their finding that loss of ZFT impairs parasite growth by showing that ZFT depletion induces stage conversion and leads to defects in both the apicoplast and mitochondrion.

Weaknesses:

(1) Excess zinc was shown not to alter ZFT expression, but a cation chelator (TPEN) did lead to decreased expression. While TPEN is often used to reduce zinc levels, does it have any effect on iron levels? Could the reduction in ZFT after TPEN treatment be due to a reduction in the level of iron or another cation?

(2) ZFT expression was found to be dynamic depending on the size of the vacuole, based on mean fluorescence intensity measurements. Looking at protein levels by Western blot at different times during infection would strengthen this finding.

(3) ZFT localization remained at the parasite periphery under low iron conditions. However, in the images shown in Figure S1c, larger vacuoles (containing 4-8 parasites) are shown for the untreated conditions, and single parasite-containing vacuoles are shown for the low iron condition. As ZFT localization is predominantly at the basal end of the parasite in larger PV and at the parasite periphery for smaller vacuoles, it would be better to compare vacuoles of similar size between the untreated and low-iron conditions.

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Reviewer #3 (Public review):

Summary:

Aghabi et al set out to characterize a *T. gondii* transmembrane protein with a ZIP domain, termed ZFT. The authors investigate the consequences of ZFT downregulation and overexpression for parasite fitness. Downregulation of ZFT causes defects in the parasite's endosymbiotic organelles, the apicoplast and the mitochondrion. Specifically, lack of ZFT causes a decrease in mitochondrial respiration, consistent with its role as an iron transporter. This impact on the mitochondria appears to trigger partial differentiation to bradyzoites. The authors furthermore demonstrate that expression of TgZFT can rescue a yeast mutant lacking its zinc transporter and perform an array of direct metal ion measurements, including X-ray fluorescence microscopy and inductively coupled mass spectrometry (ICP-MS). These reveal reduced metal ions in parasites depleted in ZFT. Overall, the data by Aghabi et al. reveal that ZFT is a major metal ion transporter in *T. gondii*, importing iron and zinc for diverse essential processes.

Strengths:

This study's strength lies in the thorough characterization of the transporter. The authors combine a number of techniques to measure the impact of ZFT depletion, ranging from the direct measurement of metal ions to determining the consequences for the parasite's metabolism (mitochondrial respiration), as well as performing a yeast mutant complementation. This work is very thorough and clearly presented, leaving little doubt about this protein's function.

Weaknesses:

This study offers no major novel insights into the biology of *T. gondii*. The transporter was already annotated as a zinc transporter (ToxoDB), was deemed essential (PMID: 27594426), and localized to the plasma membrane (PMID: 33053376). This study mostly confirms and validates these previous datasets. The authors identify three other proteins with a ZIT domain. Particularly, the role of TGME49_225530 is intriguing, as it is likely fitness-conferring (score: -2.8, PMID: 27594426) and has no subcellular localization assigned. Characterizing this protein as well, revealing its localization, and identifying if and how these transporters coordinate metal ion transport would have been worthwhile.

Another weakness is the data related to the impact of ZFT downregulation on the apicoplast in Figure 4. The authors show that downregulation of ZFT causes an increase in elongated apicoplasts (Figure 4d). The subsequent panels seem to show that the parasites present a dramatic growth defect at that time point. This growth arrest can directly explain the elongated apicoplast, but does not allow any conclusion about an impact on the organelle. In any case, an assessment of 'delayed death' as presented in Figure 4c seems futile, since the many other processes affected by zinc and iron depletion likely cause a rapid death, masking any potential delayed death.

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